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Ko

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(54) **METHOD AND DEVICE FOR SENSING FOR
SL PRS RESOURCE SELECTION**

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(KR)

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(KR)

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U.S.C. 154(b) by 378 days.

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(57) **ABSTRACT**

A method of operating a first device **100** in a wireless
communication system is proposed. The method may com-
prise: determining at least one first SL PRS candidate
resource for transmitting a first PRS; receiving, from a
second device **200**, first control information related to SL
PRS through a first control channel related to SL PRS;
obtaining information related to at least one first SL PRS
resource to be used by the second device **200**, based on the
first control information; excluding at least one second SL
PRS resource from the at least one first SL PRS candidate
resource, based on an RSRP value related to the first control
information being greater than or equal to a first threshold
and a number of the at least one second SL PRS resource,
in which the at least one first SL PRS candidate resource and
the at least one first SL PRS resource overlap, being greater
than or equal to a second threshold.

Related U.S. Application Data

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22, 2022, provisional application No. 63/315,471,
filed on Mar. 1, 2022.

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H04W 64/00 (2009.01)
H04W 92/18 (2009.01)

(52) **U.S. Cl.**
CPC **H04W 64/00** (2013.01); **H04W 92/18**
(2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

15 Claims, 22 Drawing Sheets

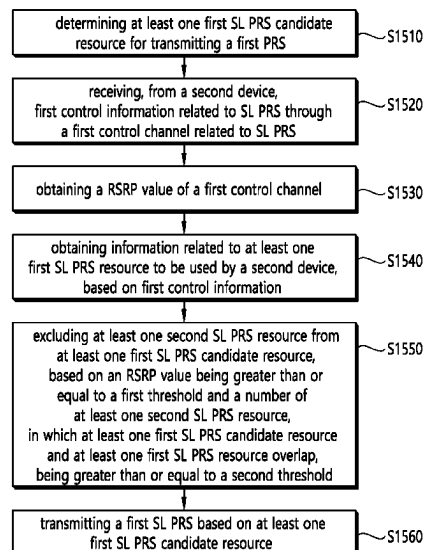


FIG. 1

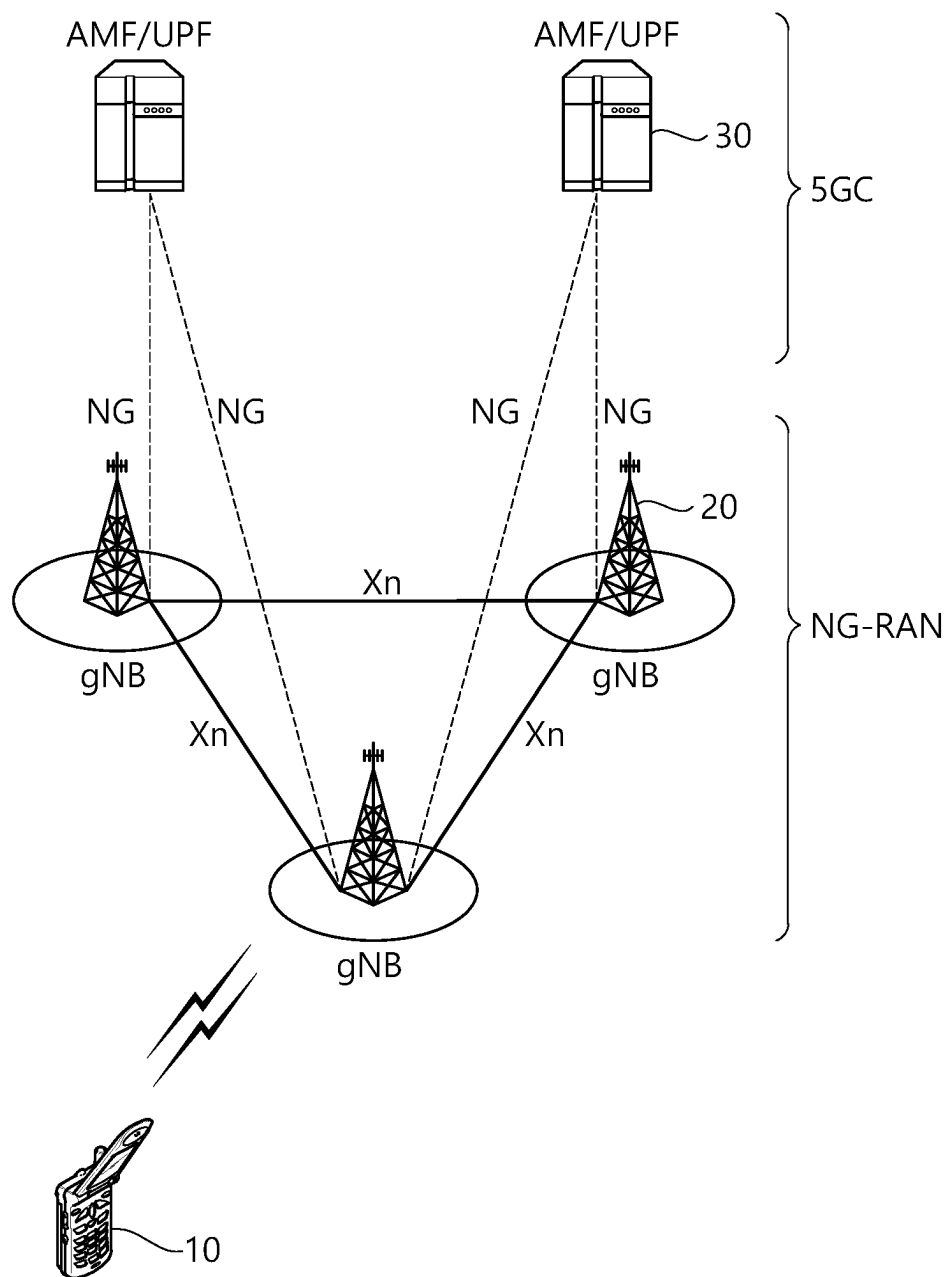


FIG. 2

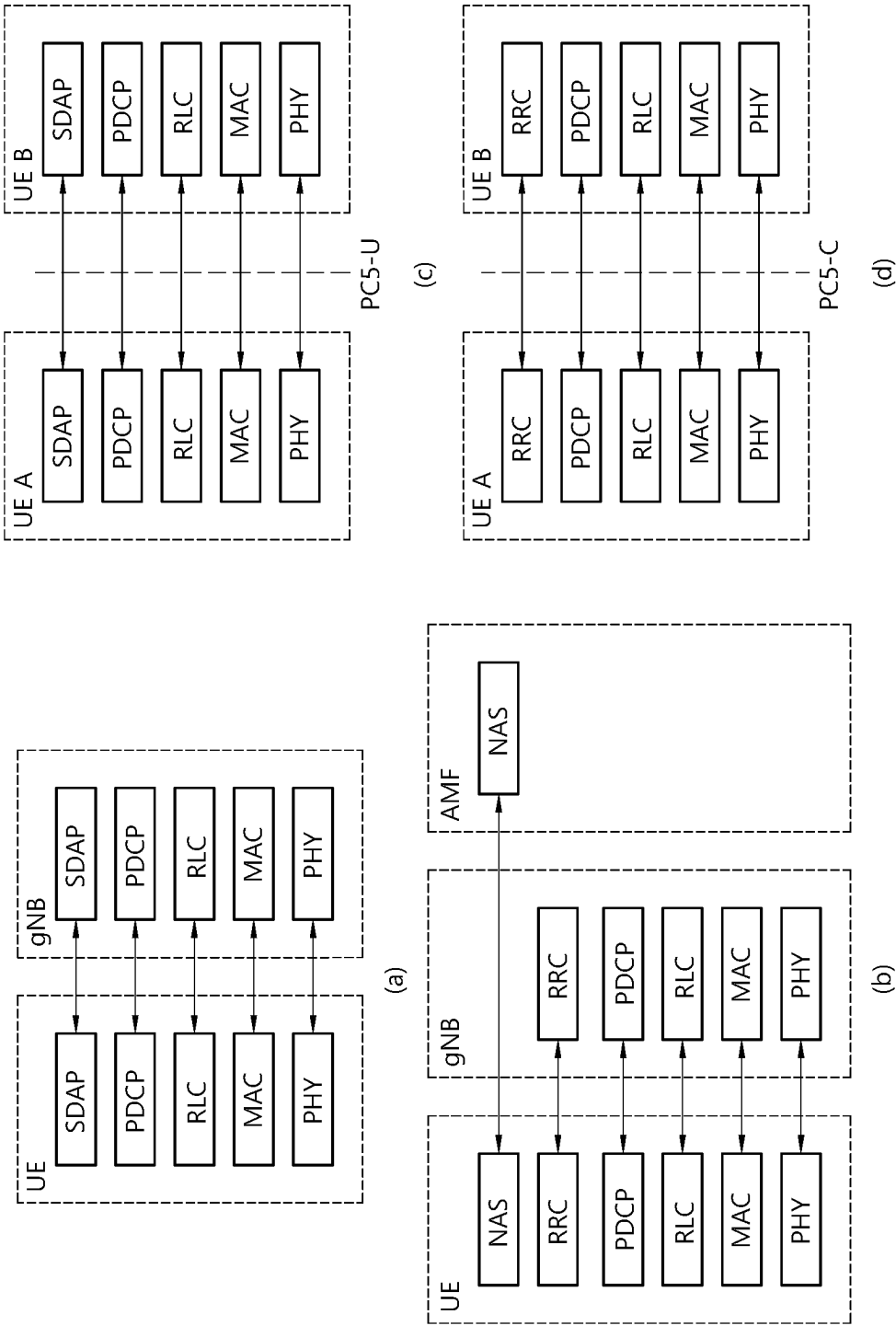


FIG. 3

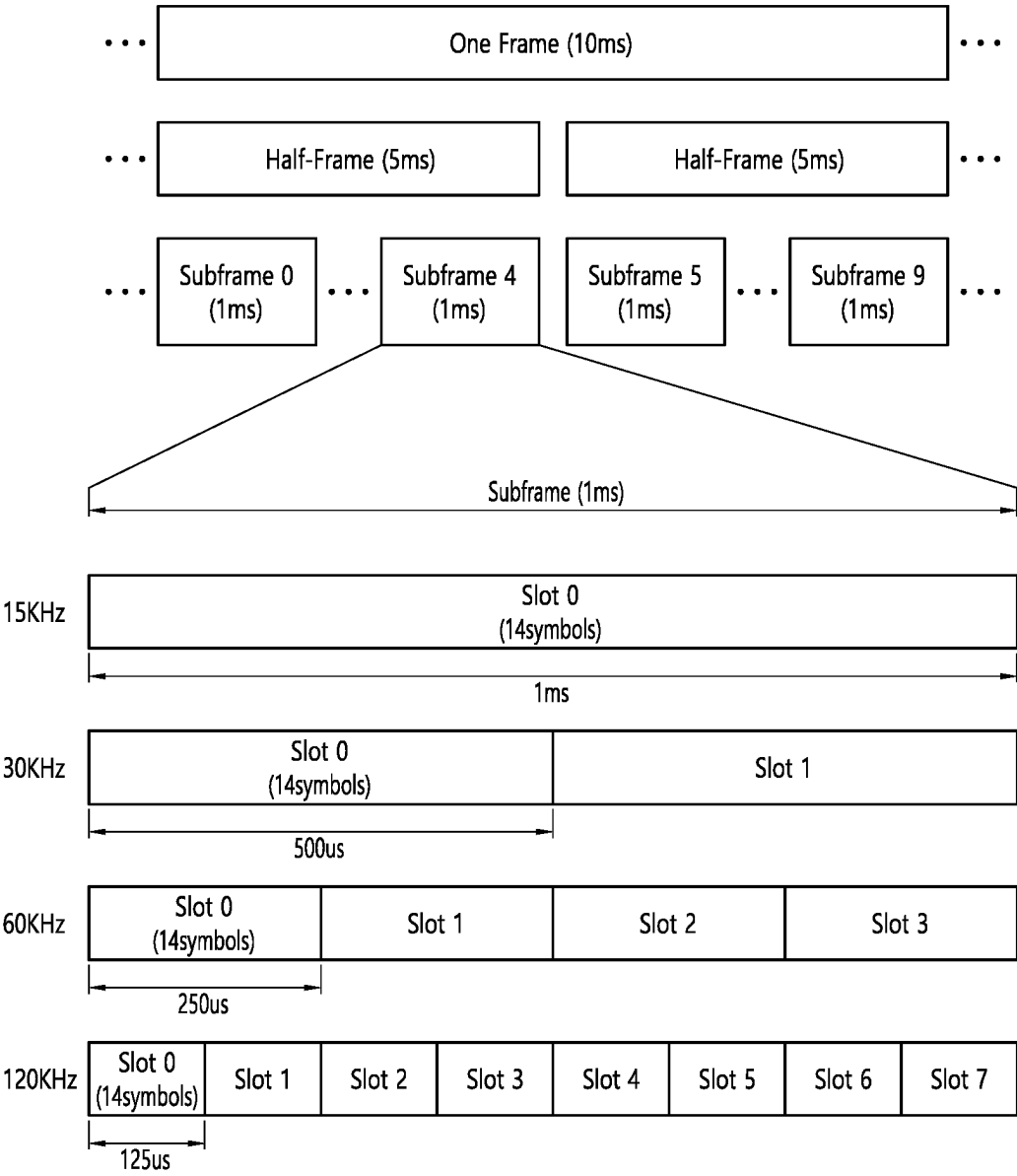


FIG. 4

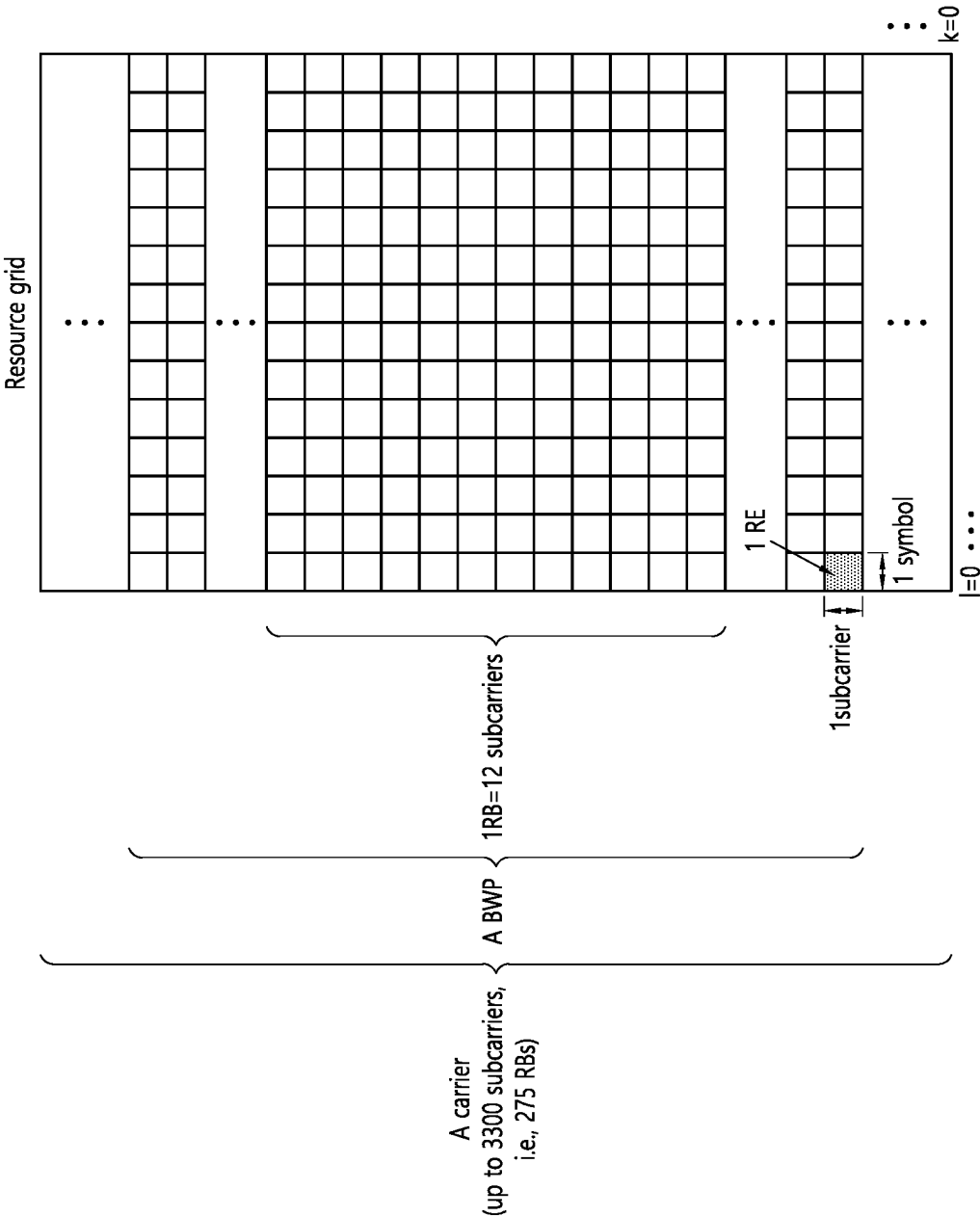


FIG. 5

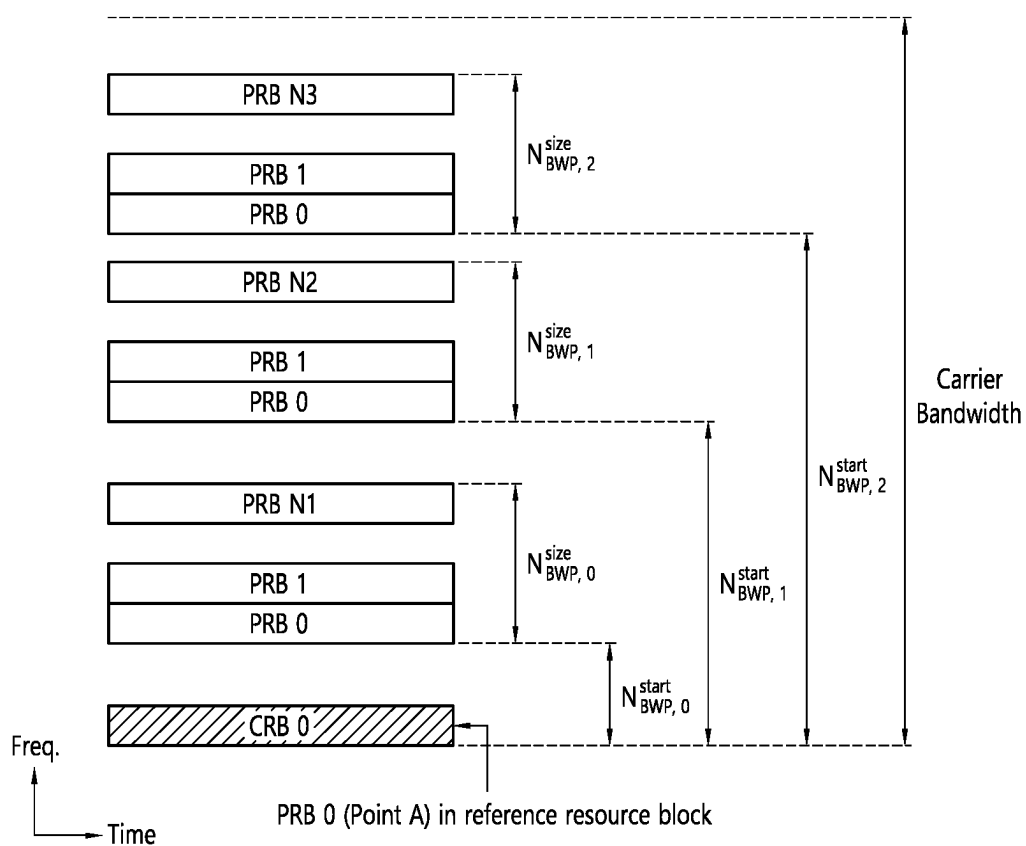


FIG. 6

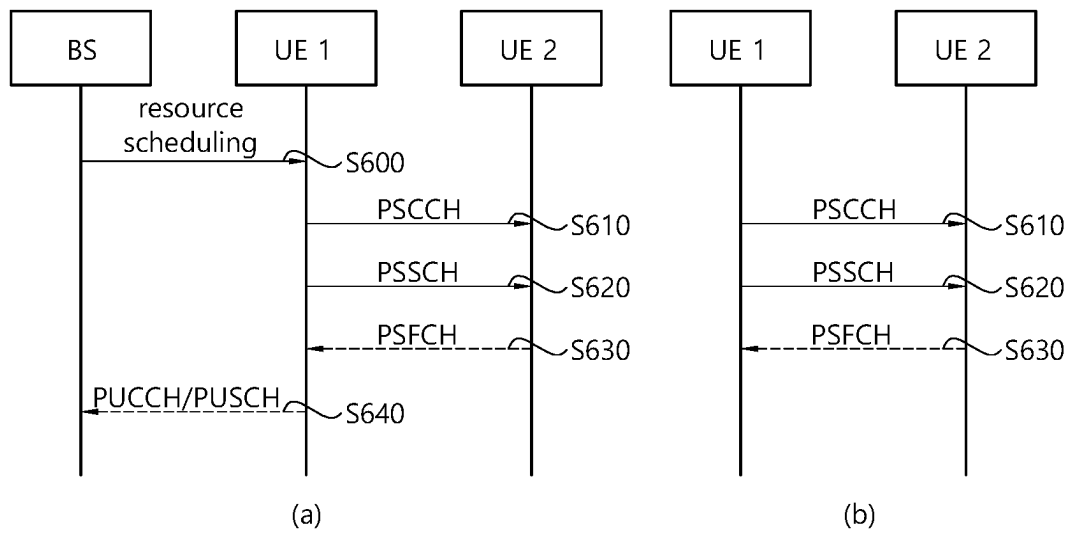


FIG. 7

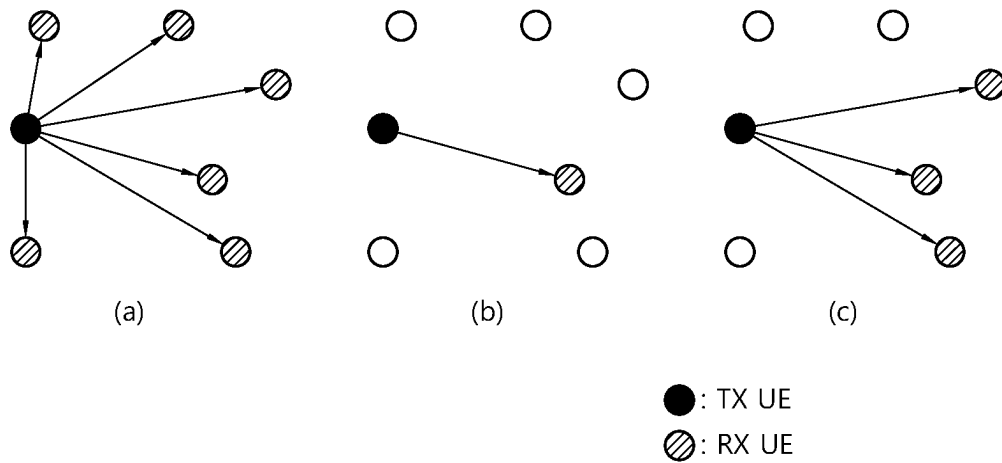


FIG. 8

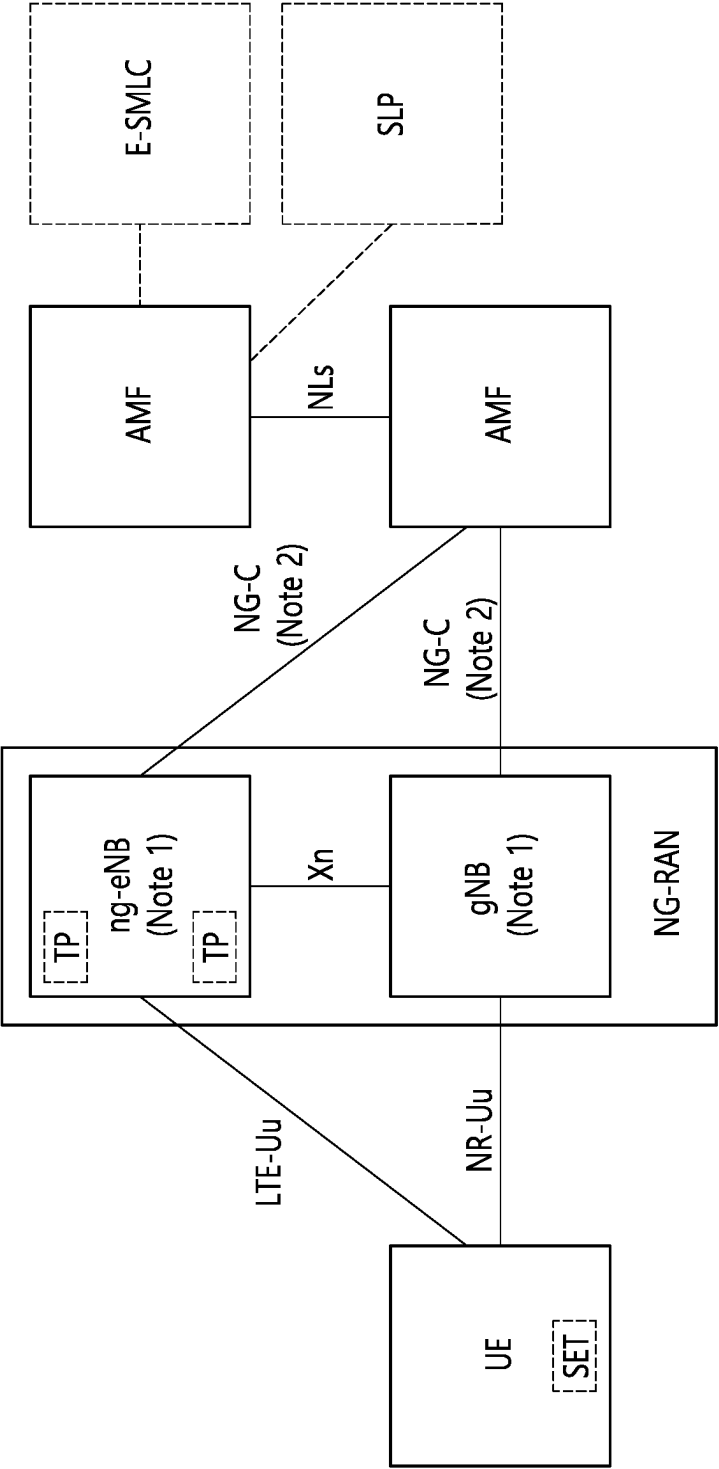


FIG. 9

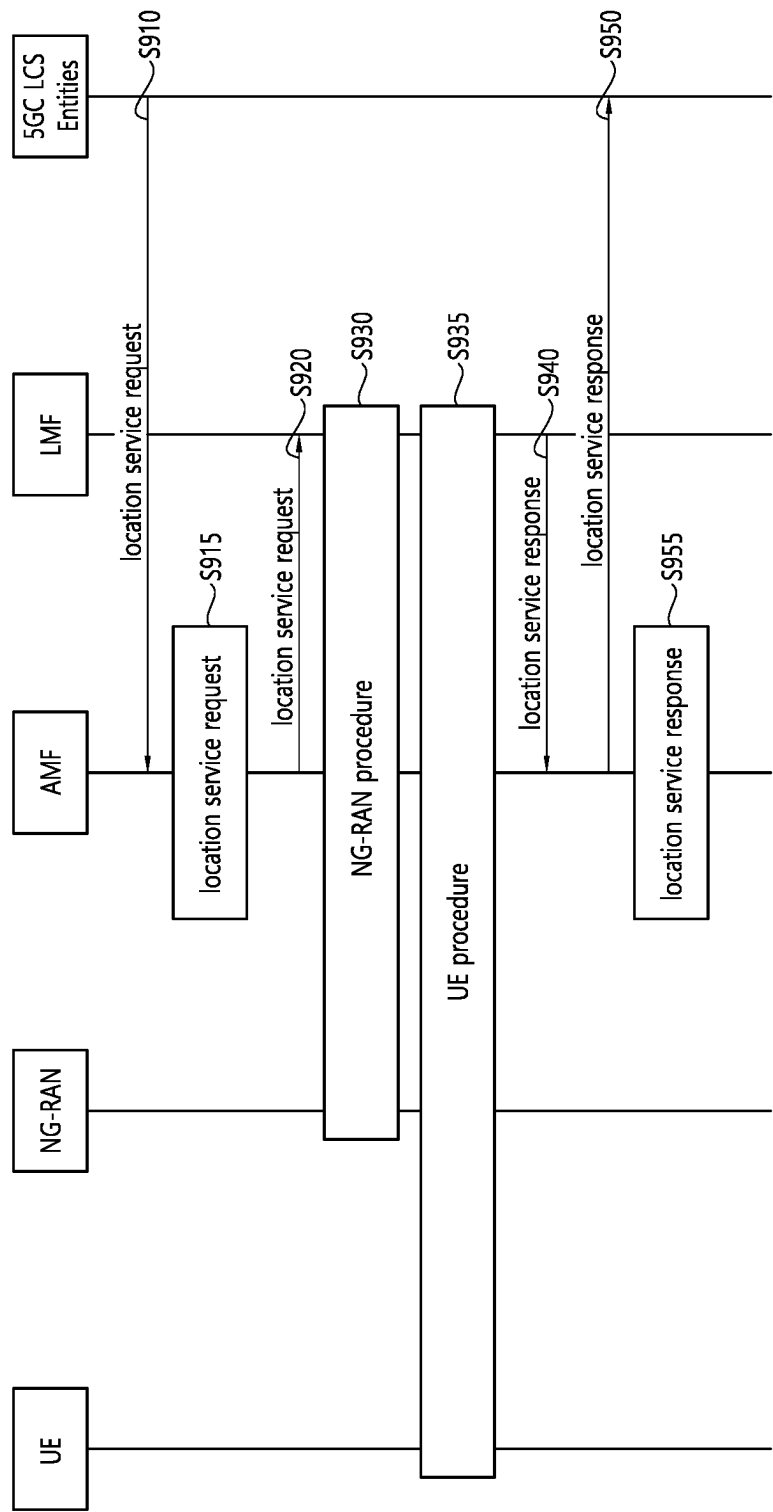


FIG. 10

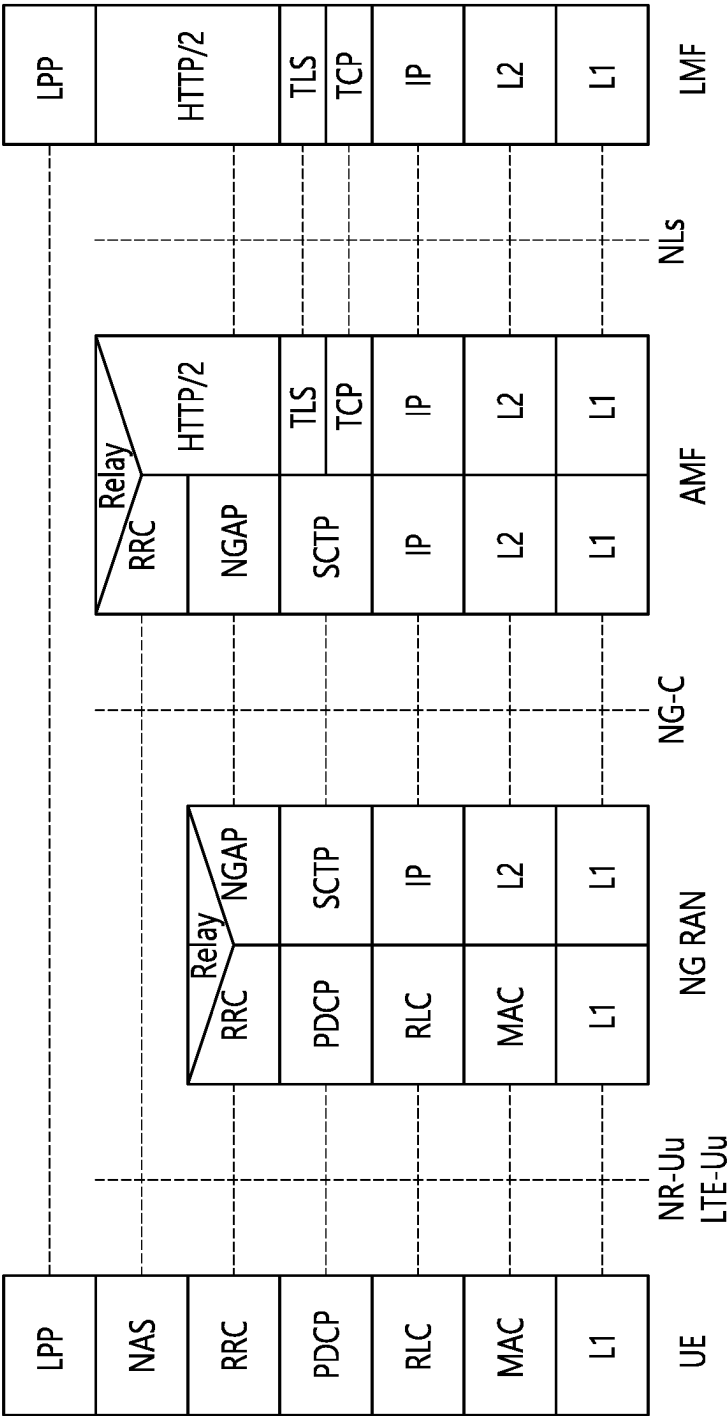


FIG. 11

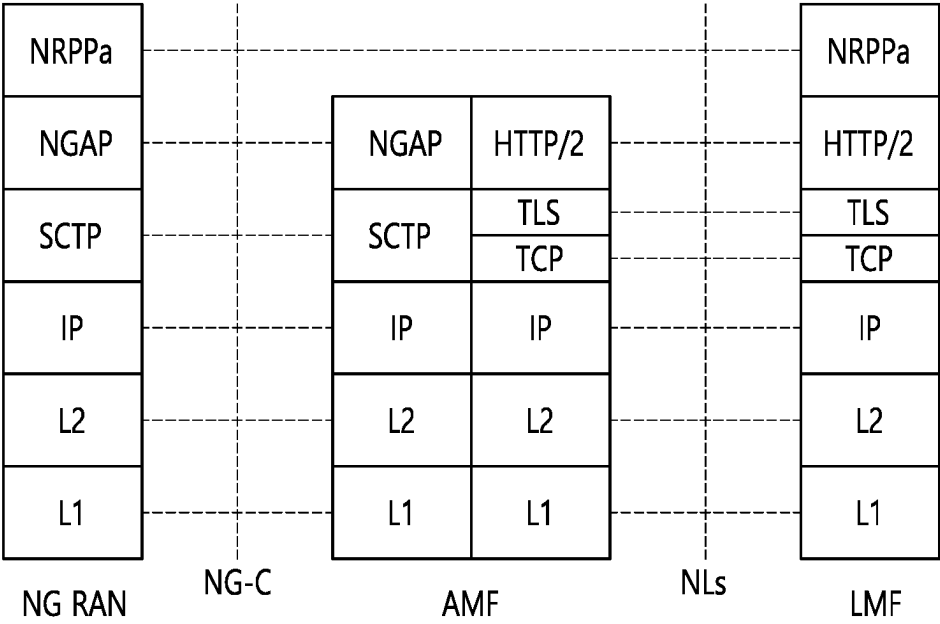


FIG. 12

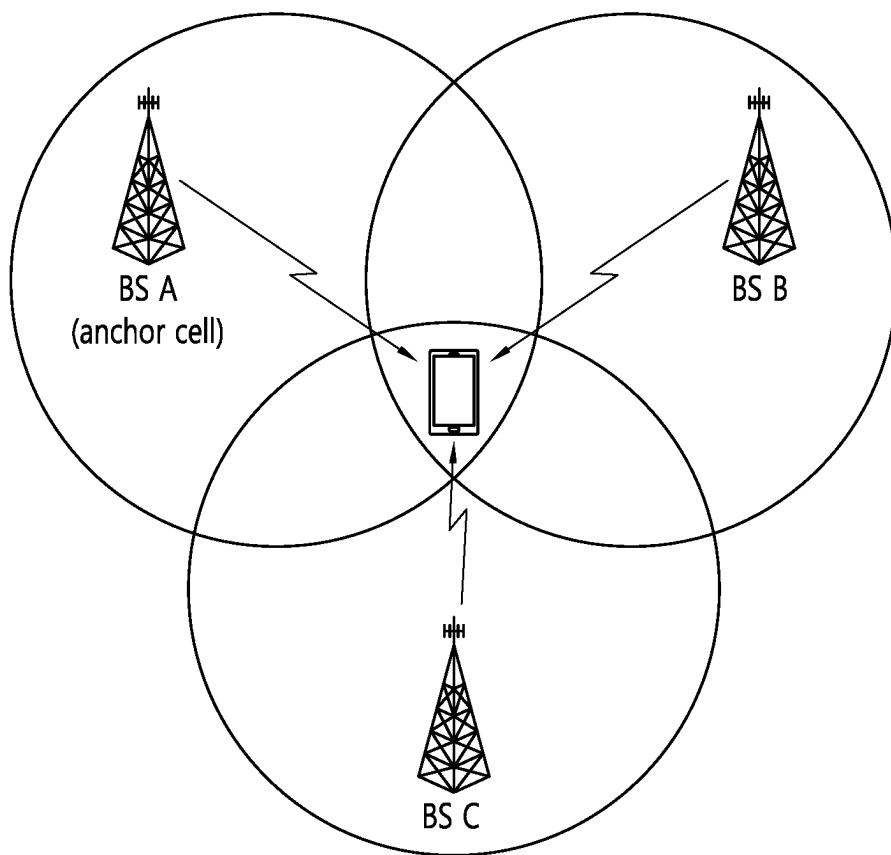


FIG. 13

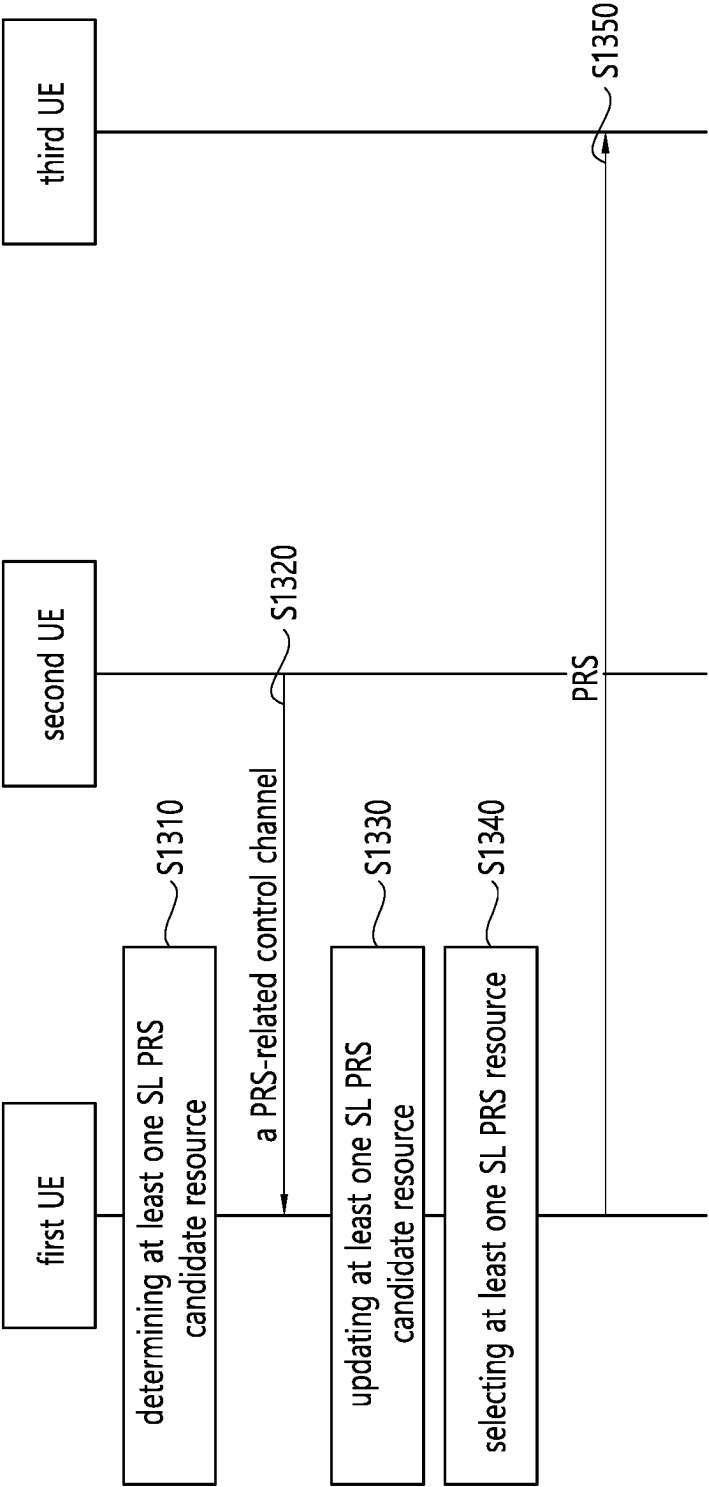


FIG. 14

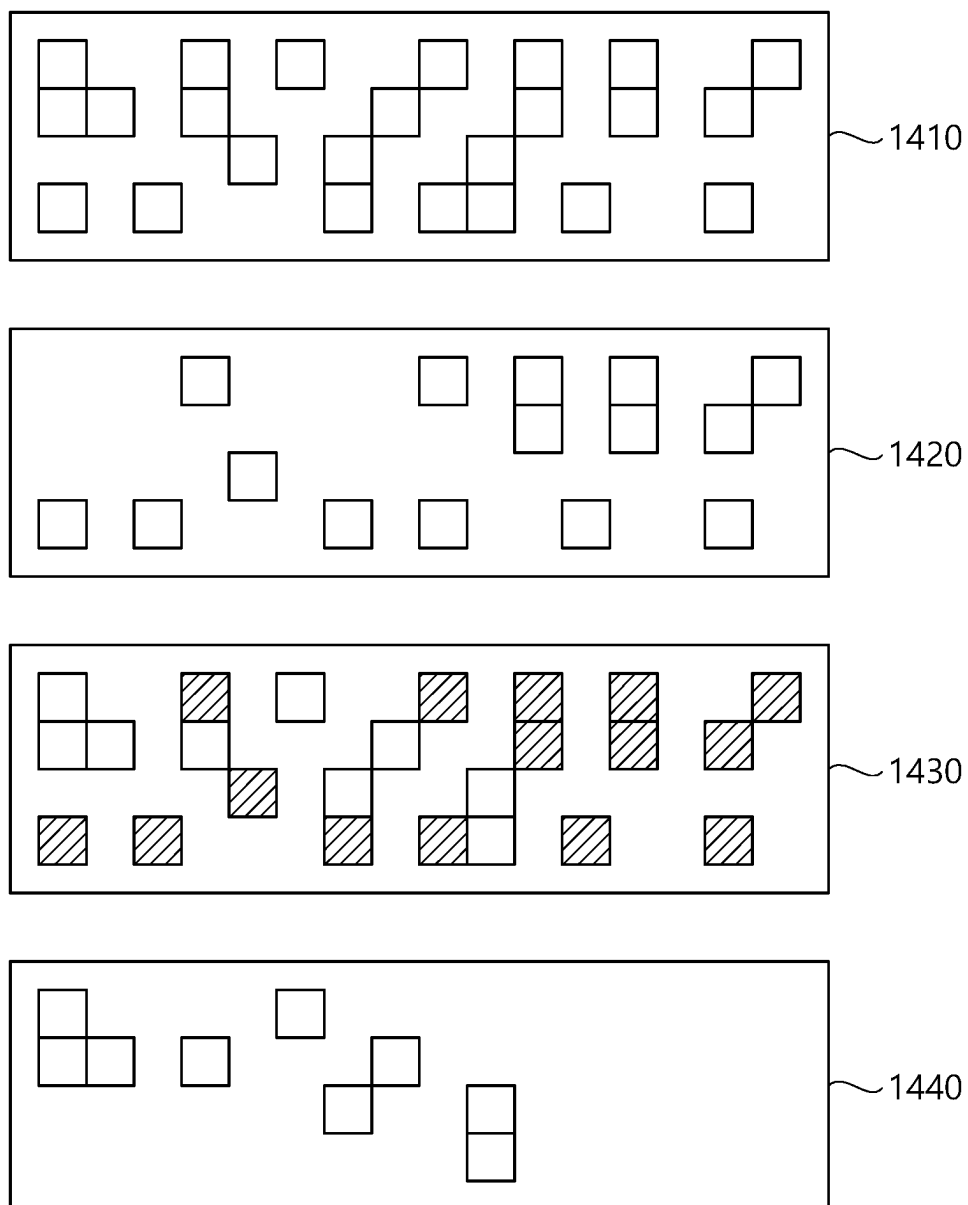


FIG. 15

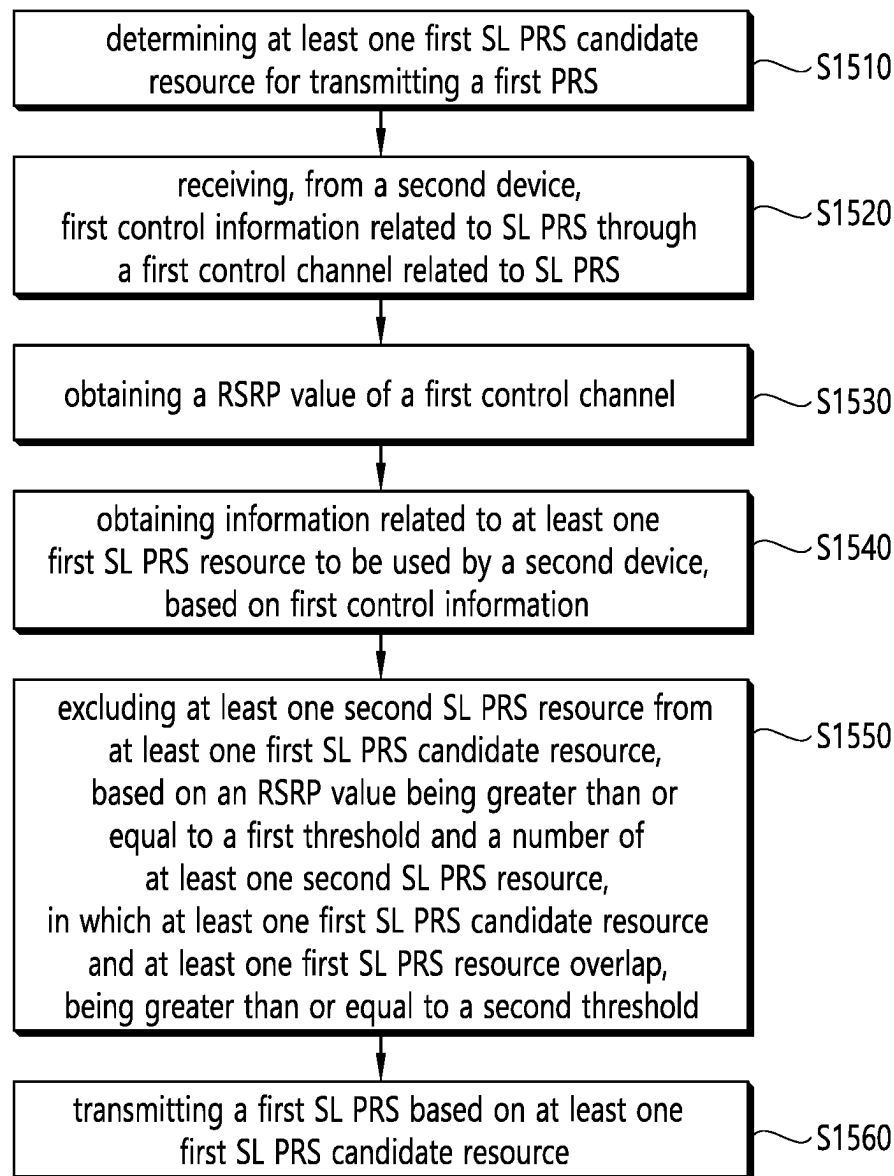


FIG. 16

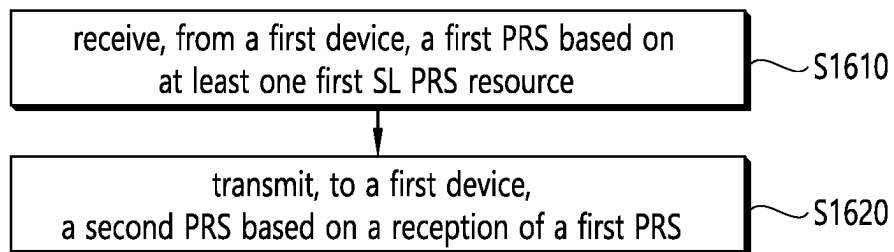


FIG. 17

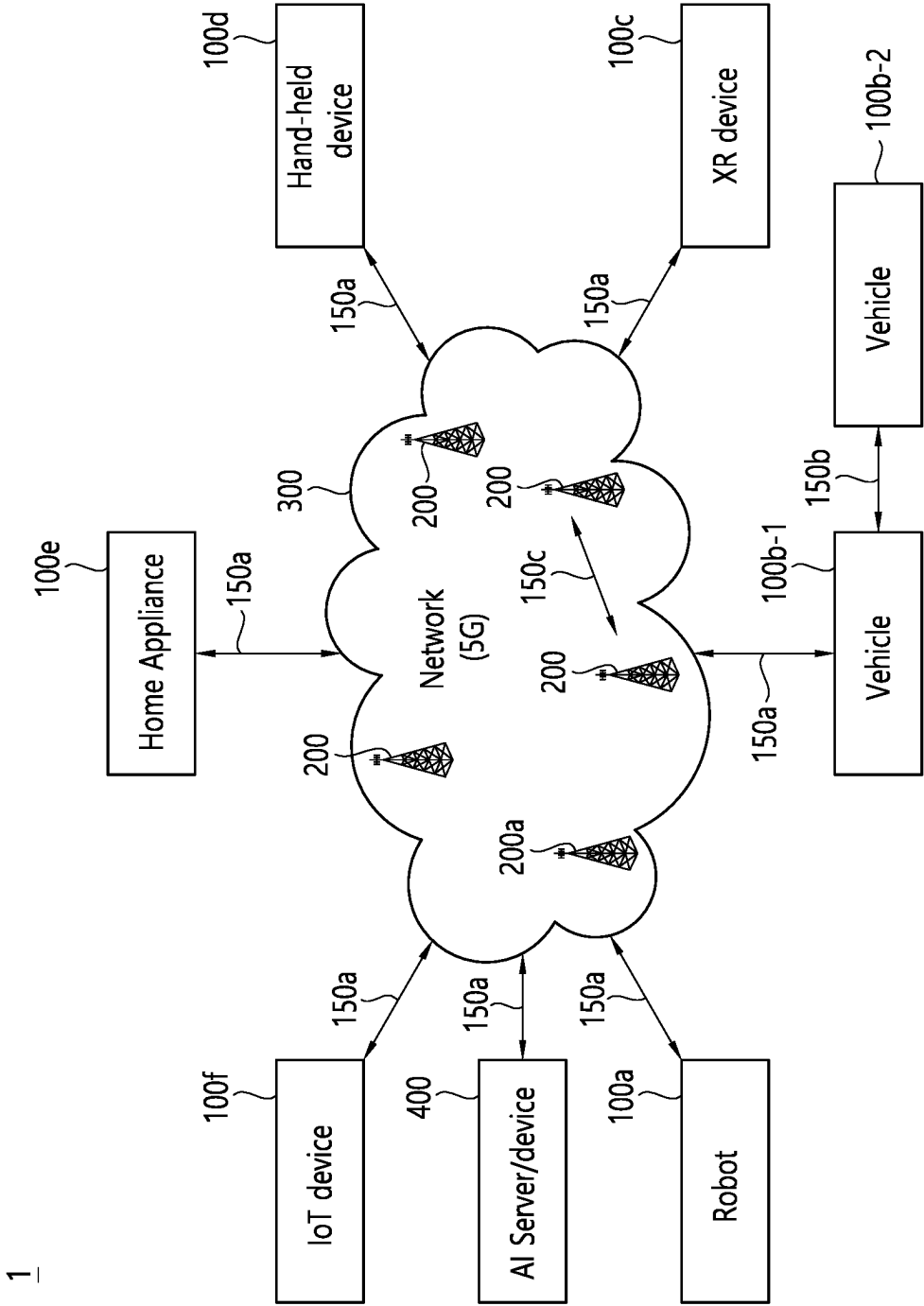


FIG. 18

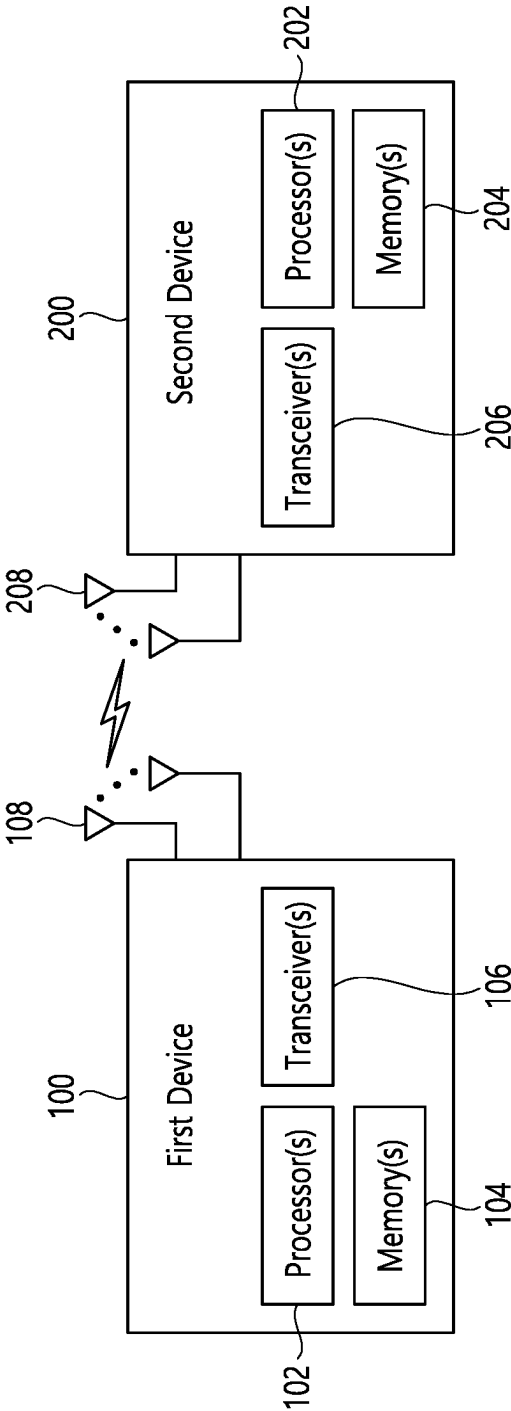


FIG. 19

1000(102/106, 202/206)

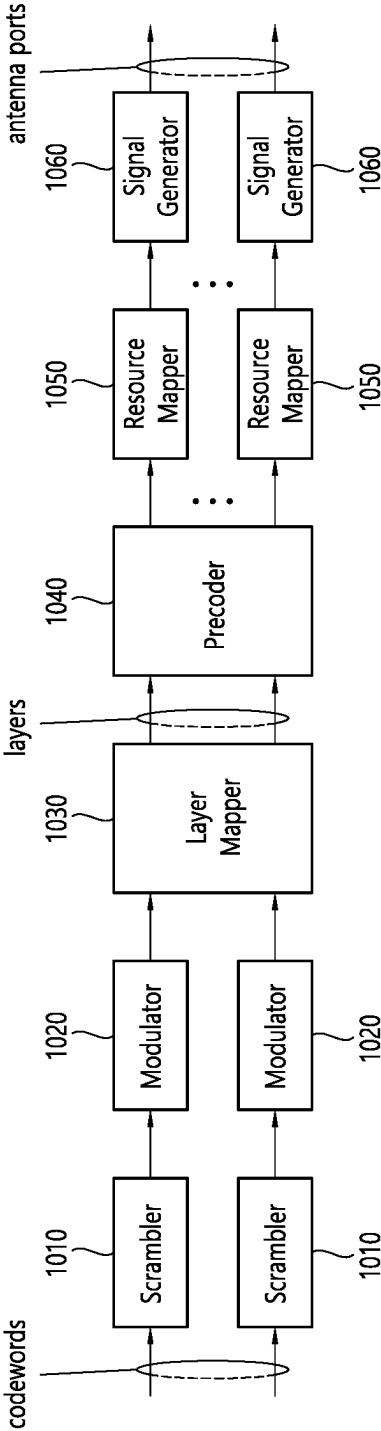


FIG. 20

Device (100,200)

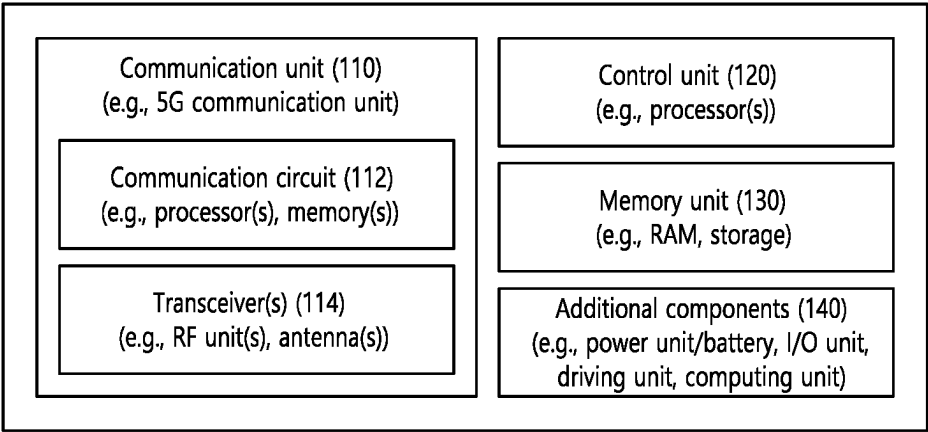


FIG. 21

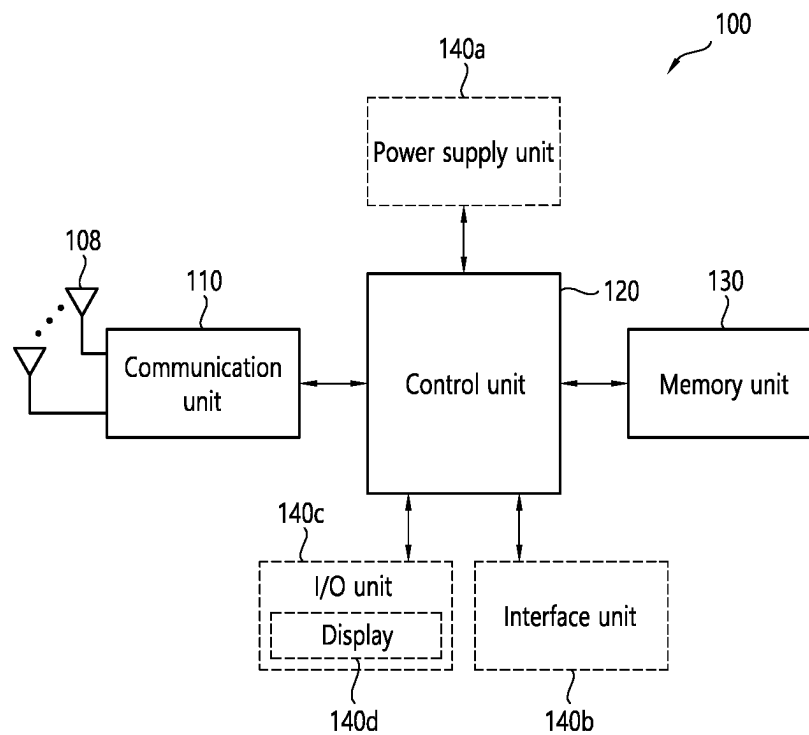
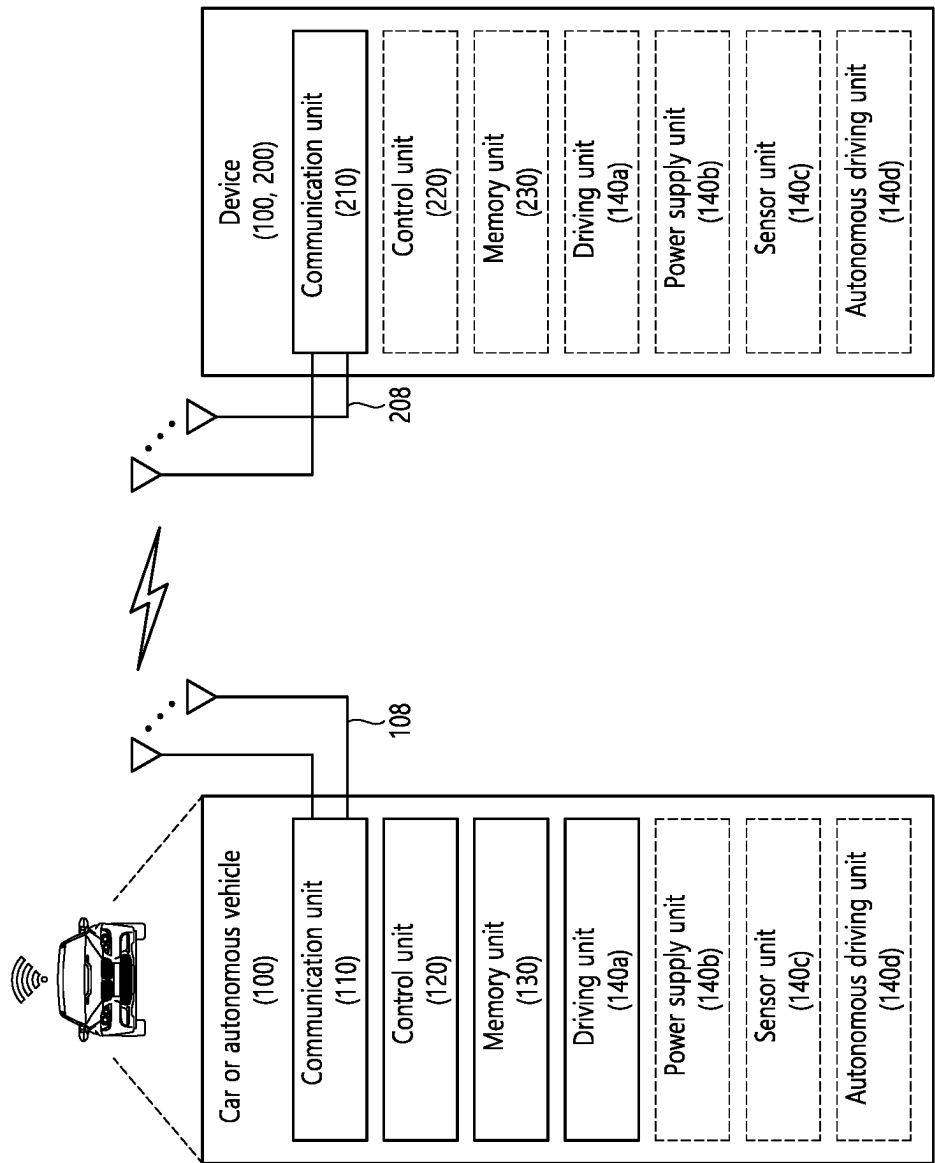


FIG. 22



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METHOD AND DEVICE FOR SENSING FOR SL PRS RESOURCE SELECTION

CROSS-REFERENCE TO RELATED APPLICATION(S)

Pursuant to 35 U.S.C. § 119(e), this application claims the benefit of U.S. Provisional Patent Application Nos. 63/315,471, filed on Mar. 1, 2022, and 63/333,619, filed on Apr. 22, 2022, the contents of which are all hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

This disclosure relates to a wireless communication system.

BACKGROUND

Sidelink (SL) communication is a communication scheme in which a direct link is established between User Equipments (UEs) and the UEs exchange voice and data directly with each other without intervention of an evolved Node B (eNB). SL communication is under consideration as a solution to the overhead of an eNB caused by rapidly increasing data traffic. Vehicle-to-everything (V2X) refers to a communication technology through which a vehicle exchanges information with another vehicle, a pedestrian, an object having an infrastructure (or infra) established therein, and so on. The V2X may be divided into 4 types, such as vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). The V2X communication may be provided via a PC5 interface and/or Uu interface.

Meanwhile, as a wider range of communication devices require larger communication capacities, the need for mobile broadband communication that is more enhanced than the existing Radio Access Technology (RAT) is rising. Accordingly, discussions are made on services and user equipment (UE) that are sensitive to reliability and latency. And, a next generation radio access technology that is based on the enhanced mobile broadband communication, massive Machine Type Communication (MTC), Ultra-Reliable and Low Latency Communication (URLLC), and so on, may be referred to as a new radio access technology (RAT) or new radio (NR). Herein, the NR may also support vehicle-to-everything (V2X) communication.

SUMMARY

According to an embodiment of the present disclosure, a method for performing, by a first device, wireless communication may be proposed. For example, the method may comprise: determining at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receiving, from a second device, first control information related to SL PRS through a first control channel related to SL PRS; obtaining a reference signal received power (RSRP) value of the first control channel; obtaining information related to at least one first SL PRS resource to be used by the second device, based on the first control information; excluding at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap,

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being greater than or equal to a second threshold; and transmitting the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a first device for performing wireless communication may be proposed. For example, the first device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers. For example, the one or more processors may execute the instructions to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a device adapted to control a first user equipment (UE), the device comprising: one or more processors; and one or more memories operably connectable to the one or more processors and storing instructions may be proposed. For example, the one or more processors may execute the instructions to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second UE, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second UE, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a non-transitory computer-readable storage medium storing instructions may be proposed. For example, the instructions, when executed, may cause a first device to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second

threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a method for performing, by a second device, wireless communication may be proposed. For example, the method may comprise: receiving, from a first device, a first positioning reference signal (PRS) based on at least one first sidelink (SL) PRS resource; and transmitting, to the first device, a second PRS based on the reception of the first PRS, wherein first control information related to PRS may be received to the first device through a first control channel related to PRS, wherein a third SL PRS candidate resource, based on an RSRP value of the first control channel being greater than or equal to a first threshold and a number of the at least one third SL PRS resource, in which the at least one first SL PRS candidate resource and at least one second SL PRS resource overlap, being greater than or equal to a second threshold, wherein information related to the at least one second SL PRS resource may be obtained by the first device based on the first control information, wherein the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, and wherein positioning for the first device may be performed based on the first PRS and the second PRS.

According to an embodiment of the present disclosure, a second device for performing wireless communication may be proposed. For example, the second device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers, wherein the one or more processors execute the instructions to: receive, from a first device, a first positioning reference signal (PRS) based on at least one first sidelink (SL) PRS resource; and transmit, to the first device, a second PRS based on the reception of the first PRS, wherein first control information related to PRS may be received to the first device through a first control channel related to PRS, wherein a third SL PRS candidate resource may be excluded from at least one first SL PRS candidate resource, based on an RSRP value of the first control channel being greater than or equal to a first threshold and a number of the at least one third SL PRS resource, in which the at least one first SL PRS candidate resource and at least one second SL PRS resource overlap, being greater than or equal to a second threshold, wherein information related to the at least one second SL PRS resource may be obtained by the first device based on the first control information, wherein the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, and wherein positioning for the first device may be performed based on the first PRS and the second PRS.

A UE may efficiently perform sidelink communication.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a structure of an NR system, based on an embodiment of the present disclosure.

FIG. 2 shows a radio protocol architecture, based on an embodiment of the present disclosure.

FIG. 3 shows a structure of a radio frame of an NR, based on an embodiment of the present disclosure.

FIG. 4 shows a structure of a slot of an NR frame, based on an embodiment of the present disclosure.

FIG. 5 shows an example of a BWP, based on an embodiment of the present disclosure.

FIG. 6 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, based on an embodiment of the present disclosure.

FIG. 7 shows three cast types, based on an embodiment of the present disclosure.

FIG. 8 shows an example of an architecture in a 5G system in which positioning for a UE connected to a Next Generation-Radio Access Network (NG-RAN) or E-UTRAN is possible, according to an embodiment of the present disclosure.

FIG. 9 shows an implementation example of a network for measuring a position of a UE, according to an embodiment of the present disclosure.

FIG. 10 shows an example of a protocol layer used to support LTE Positioning Protocol (LPP) message transmission between an LMF and a UE, according to an embodiment of the present disclosure.

FIG. 11 shows an example of a protocol layer used to support NR Positioning Protocol A (NRPPa) PDU transmission between an LMF and an NG-RAN node, according to an embodiment of the present disclosure.

FIG. 12 shows an Observed Time Difference Of Arrival (OTDOA) positioning method according to an embodiment of the present disclosure.

FIG. 13 shows a procedure for a first UE to transmit an SL PRS based on at least one SL PRS resource according to an embodiment of the present disclosure.

FIG. 14 shows an embodiment in which SL PRS candidate resources are updated according to an embodiment of the present disclosure.

FIG. 15 shows a procedure for performing wireless communication by a first device according to an embodiment of the present disclosure.

FIG. 16 shows a procedure for performing wireless communication by a second device according to an embodiment of the present disclosure.

FIG. 17 shows a communication system 1, based on an embodiment of the present disclosure.

FIG. 18 shows wireless devices, based on an embodiment of the present disclosure.

FIG. 19 shows a signal process circuit for a transmission signal, based on an embodiment of the present disclosure.

FIG. 20 shows another example of a wireless device, based on an embodiment of the present disclosure.

FIG. 21 shows a hand-held device, based on an embodiment of the present disclosure.

FIG. 22 shows a vehicle or an autonomous vehicle, based on an embodiment of the present disclosure.

DETAILED DESCRIPTION

In the present disclosure, “A or B” may mean “only A”, “only B” or “both A and B.” In other words, in the present disclosure, “A or B” may be interpreted as “A and/or B”. For example, in the present disclosure, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.

A slash (/) or comma used in the present disclosure may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.

In the present disclosure, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present disclosure, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.

In addition, in the present disclosure, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, and C”. In addition, “at least one of A, B, or C” or “at least one of A, B, and/or C” may mean “at least one of A, B, and C”.

In addition, a parenthesis used in the present disclosure may mean “for example”. Specifically, when indicated as “control information (PDCCH)”, it may mean that “PDCCH” is proposed as an example of the “control information”. In other words, the “control information” of the present disclosure is not limited to “PDCCH”, and “PDCCH” may be proposed as an example of the “control information”. In addition, when indicated as “control information (i.e., PDCCH)”, it may also mean that “PDCCH” is proposed as an example of the “control information”.

In the following description, ‘when, if, or in case of’ may be replaced with ‘based on’.

A technical feature described individually in one figure in the present disclosure may be individually implemented, or may be simultaneously implemented.

In the present disclosure, a higher layer parameter may be a parameter which is configured, pre-configured or pre-defined for a UE. For example, a base station or a network may transmit the higher layer parameter to the UE. For example, the higher layer parameter may be transmitted through radio resource control (RRC) signaling or medium access control (MAC) signaling.

The technology described below may be used in various wireless communication systems such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), single carrier frequency division multiple access (SC-FDMA), and so on. The CDMA may be implemented with a radio technology, such as universal terrestrial radio access (UTRA) or CDMA-2000. The TDMA may be implemented with a radio technology, such as global system for mobile communications (GSM)/general packet radio service (GPRS)/enhanced data rate for GSM evolution (EDGE). The OFDMA may be implemented with a radio technology, such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, evolved UTRA (E-UTRA), and so on. IEEE 802.16m is an evolved version of IEEE 802.16e and provides backward compatibility with a system based on the IEEE 802.16e. The UTRA is part of a universal mobile telecommunication system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is part of an evolved UMTS (E-UMTS) using the E-UTRA. The 3GPP LTE uses the OFDMA in a downlink and uses the SC-FDMA in an uplink. LTE-advanced (LTE-A) is an evolution of the LTE.

5G NR is a successive technology of LTE-A corresponding to a new Clean-slate type mobile communication system having the characteristics of high performance, low latency, high availability, and so on. 5G NR may use resources of all spectrum available for usage including low frequency bands of less than 1 GHz, middle frequency bands ranging from 1 GHz to 10 GHz, high frequency (millimeter waves) of 24 GHz or more, and so on.

For clarity in the description, the following description will mostly focus on LTE-A or 5G NR. However, technical features according to an embodiment of the present disclosure will not be limited only to this.

For terms and techniques not specifically described among terms and techniques used in this specification, a

wireless communication standard document published before the present specification is filed may be referred to.

FIG. 1 shows a structure of an NR system, based on an embodiment of the present disclosure. The embodiment of FIG. 1 may be combined with various embodiments of the present disclosure.

Referring to FIG. 1, a next generation-radio access network (NG-RAN) may include a BS 20 providing a UE 10 with a user plane and control plane protocol termination. For example, the BS 20 may include a next generation-Node B (gNB) and/or an evolved-NodeB (eNB). For example, the UE 10 may be fixed or mobile and may be referred to as other terms, such as a mobile station (MS), a user terminal (UT), a subscriber station (SS), a mobile terminal (MT), wireless device, and so on. For example, the BS may be referred to as a fixed station which communicates with the UE 10 and may be referred to as other terms, such as a base transceiver system (BTS), an access point (AP), and so on.

The embodiment of FIG. 1 exemplifies a case where only the gNB is included. The BSs 20 may be connected to one another via Xn interface. The BS 20 may be connected to one another via 5th generation (5G) core network (5GC) and NG interface. More specifically, the BSs 20 may be connected to an access and mobility management function (AMF) 30 via NG-C interface, and may be connected to a user plane function (UPF) 30 via NG-U interface.

Layers of a radio interface protocol between the UE and the network can be classified into a first layer (layer 1, L1), a second layer (layer 2, L2), and a third layer (layer 3, L3) based on the lower three layers of the open system interconnection (OSI) model that is well-known in the communication system. Among them, a physical (PHY) layer belonging to the first layer provides an information transfer service by using a physical channel, and a radio resource control (RRC) layer belonging to the third layer serves to control a radio resource between the UE and the network. For this, the RRC layer exchanges an RRC message between the UE and the BS.

FIG. 2 shows a radio protocol architecture, based on an embodiment of the present disclosure. The embodiment of FIG. 2 may be combined with various embodiments of the present disclosure. Specifically, (a) of FIG. 2 shows a radio protocol stack of a user plane for Uu communication, and (b) of FIG. 2 shows a radio protocol stack of a control plane for Uu communication. (c) of FIG. 2 shows a radio protocol stack of a user plane for SL communication, and (d) of FIG. 2 shows a radio protocol stack of a control plane for SL communication.

Referring to FIG. 2, a physical layer provides an upper layer with an information transfer service through a physical channel. The physical layer is connected to a medium access control (MAC) layer which is an upper layer of the physical layer through a transport channel. Data is transferred between the MAC layer and the physical layer through the transport channel. The transport channel is classified according to how and with what characteristics data is transmitted through a radio interface.

Between different physical layers, i.e., a physical layer of a transmitter and a physical layer of a receiver, data are transferred through the physical channel. The physical channel is modulated using an orthogonal frequency division multiplexing (OFDM) scheme, and utilizes time and frequency as a radio resource.

The MAC layer provides services to a radio link control (RLC) layer, which is a higher layer of the MAC layer, via a logical channel. The MAC layer provides a function of mapping multiple logical channels to multiple transport

channels. The MAC layer also provides a function of logical channel multiplexing by mapping multiple logical channels to a single transport channel. The MAC layer provides data transfer services over logical channels.

The RLC layer performs concatenation, segmentation, and reassembly of Radio Link Control Service Data Unit (RLC SDU). In order to ensure diverse quality of service (QoS) required by a radio bearer (RB), the RLC layer provides three types of operation modes, i.e., a transparent mode (TM), an unacknowledged mode (UM), and an acknowledged mode (AM). An AM RLC provides error correction through an automatic repeat request (ARQ).

A radio resource control (RRC) layer is defined only in the control plane. The RRC layer serves to control the logical channel, the transport channel, and the physical channel in association with configuration, reconfiguration and release of RBs. The RB is a logical path provided by the first layer (i.e., the physical layer or the PHY layer) and the second layer (i.e., a MAC layer, an RLC layer, a packet data convergence protocol (PDCP) layer, and a service data adaptation protocol (SDAP) layer) for data delivery between the UE and the network.

Functions of a packet data convergence protocol (PDCP) layer in the user plane include user data delivery, header compression, and ciphering. Functions of a PDCP layer in the control plane include control-plane data delivery and ciphering/integrity protection.

A service data adaptation protocol (SDAP) layer is defined only in a user plane. The SDAP layer performs mapping between a Quality of Service (QoS) flow and a data radio bearer (DRB) and QoS flow ID (QFI) marking in both DL and UL packets.

The configuration of the RB implies a process for specifying a radio protocol layer and channel properties to provide a particular service and for determining respective detailed parameters and operations. The RB can be classified into two types, i.e., a signaling RB (SRB) and a data RB (DRB). The SRB is used as a path for transmitting an RRC message in the control plane. The DRB is used as a path for transmitting user data in the user plane.

When an RRC connection is established between an RRC layer of the UE and an RRC layer of the E-UTRAN, the UE is in an RRC_CONNECTED state, and, otherwise, the UE may be in an RRC_IDLE state. In case of the NR, an RRC_INACTIVE state is additionally defined, and a UE being in the RRC_INACTIVE state may maintain its connection with a core network whereas its connection with the BS is released.

Data is transmitted from the network to the UE through a downlink transport channel. Examples of the downlink transport channel include a broadcast channel (BCH) for transmitting system information and a downlink-shared channel (SCH) for transmitting user traffic or control messages. Traffic of downlink multicast or broadcast services or the control messages can be transmitted on the downlink-SCH or an additional downlink multicast channel (MCH). Data is transmitted from the UE to the network through an uplink transport channel. Examples of the uplink transport channel include a random access channel (RACH) for transmitting an initial control message and an uplink SCH for transmitting user traffic or control messages.

Examples of logical channels belonging to a higher channel of the transport channel and mapped onto the transport channels include a broadcast channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (MCCH), a multicast traffic channel (MTCH), etc.

FIG. 3 shows a structure of a radio frame of an NR, based on an embodiment of the present disclosure. The embodiment of FIG. 3 may be combined with various embodiments of the present disclosure.

Referring to FIG. 3, in the NR, a radio frame may be used for performing uplink and downlink transmission. A radio frame has a length of 10 ms and may be defined to be configured of two half-frames (HFs). A half-frame may include five 1 ms subframes (SFs). A subframe (SF) may be divided into one or more slots, and the number of slots within a subframe may be determined based on subcarrier spacing (SCS). Each slot may include 12 or 14 OFDM(A) symbols according to a cyclic prefix (CP).

In case of using a normal CP, each slot may include 14 symbols. In case of using an extended CP, each slot may include 12 symbols. Herein, a symbol may include an OFDM symbol (or CP-OFDM symbol) and a Single Carrier-FDMA (SC-FDMA) symbol (or Discrete Fourier Transform-spread-OFDM (DFT-s-OFDM) symbol).

Table 1 shown below represents an example of a number of symbols per slot (N_{slot}^{symbol}), a number slots per frame ($N_{frame,u}^{slot}$), and a number of slots per subframe ($N_{subframe,u}^{slot}$) based on an SCS configuration (u), in a case where a normal CP is used.

TABLE 1

SCS ($15 \cdot 2^u$)	N_{slot}^{symbol}	$N_{frame,u}^{slot}$	$N_{subframe,u}^{slot}$
15 KHz (u = 0)	14	10	1
30 KHz (u = 1)	14	20	2
60 KHz (u = 2)	14	40	4
120 KHz (u = 3)	14	80	8
240 KHz (u = 4)	14	160	16

Table 2 shows an example of a number of symbols per slot, a number of slots per frame, and a number of slots per subframe based on the SCS, in a case where an extended CP is used.

TABLE 2

SCS ($15 \cdot 2^u$)	N_{slot}^{symbol}	$N_{frame,u}^{slot}$	$N_{subframe,u}^{slot}$
60 KHz (u = 2)	12	40	4

In an NR system, OFDM(A) numerologies (e.g., SCS, CP length, and so on) between multiple cells being integrate to one UE may be differently configured. Accordingly, a (absolute time) duration (or section) of a time resource (e.g., subframe, slot or TTI) (collectively referred to as a time unit (TU) for simplicity) being configured of the same number of symbols may be differently configured in the integrated cells.

In the NR, multiple numerologies or SCSs for supporting diverse 5G services may be supported. For example, in case an SCS is 15 kHz, a wide area of the conventional cellular bands may be supported, and, in case an SCS is 30 kHz/60 kHz a dense-urban, lower latency, wider carrier bandwidth may be supported. In case the SCS is 60 kHz or higher, a bandwidth that is greater than 24.25 GHz may be used in order to overcome phase noise.

An NR frequency band may be defined as two different types of frequency ranges. The two different types of frequency ranges may be FR1 and FR2. The values of the frequency ranges may be changed (or varied), and, for example, the two different types of frequency ranges may be as shown below in Table 3. Among the frequency ranges that

are used in an NR system, FR1 may mean a “sub 6 GHz range”, and FR2 may mean an “above 6 GHz range” and may also be referred to as a millimeter wave (mmW).

TABLE 3

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	450 MHz-6000 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

As described above, the values of the frequency ranges in the NR system may be changed (or varied). For example, as shown below in Table 4, FR1 may include a band within a range of 410 MHz to 7125 MHz. More specifically, FR1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher. For example, a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher being included in FR1 may include an unlicensed band. The unlicensed band may be used for diverse purposes, e.g., the unlicensed band for vehicle-specific communication (e.g., automated driving).

TABLE 4

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

FIG. 4 shows a structure of a slot of an NR frame, based on an embodiment of the present disclosure. The embodiment of FIG. 4 may be combined with various embodiments of the present disclosure.

Referring to FIG. 4, a slot includes a plurality of symbols in a time domain. For example, in case of a normal CP, one slot may include 14 symbols. However, in case of an extended CP, one slot may include 12 symbols. Alternatively, in case of a normal CP, one slot may include 7 symbols. However, in case of an extended CP, one slot may include 6 symbols. A carrier includes a plurality of subcarriers in a frequency domain. A Resource Block (RB) may be defined as a plurality of consecutive subcarriers (e.g., 12 subcarriers) in the frequency domain. A Bandwidth Part (BWP) may be defined as a plurality of consecutive (Physical) Resource Blocks ((P)RBs) in the frequency domain, and the BWP may correspond to one numerology (e.g., SCS, CP length, and so on).

A carrier may include a maximum of N number BWPs (e.g., 5 BWPs). Data communication may be performed via an activated BWP. Each element may be referred to as a Resource Element (RE) within a resource grid and one complex symbol may be mapped to each element.

Hereinafter, a bandwidth part (BWP) and a carrier will be described.

The BWP may be a set of consecutive physical resource blocks (PRBs) in a given numerology. The PRB may be selected from consecutive sub-sets of common resource blocks (CRBs) for the given numerology on a given carrier

For example, the BWP may be at least any one of an active BWP, an initial BWP, and/or a default BWP. For example, the UE may not monitor downlink radio link quality in a DL BWP other than an active DL BWP on a primary cell (PCell). For example, the UE may not receive PDCCH, physical downlink shared channel (PDSCH), or channel state information-reference signal (CSI-RS) (ex-

cluding RRM) outside the active DL BWP. For example, the UE may not trigger a channel state information (CSI) report for the inactive DL BWP. For example, the UE may not transmit physical uplink control channel (PUCCH) or physical uplink shared channel (PUSCH) outside an active UL BWP. For example, in a downlink case, the initial BWP may be given as a consecutive RB set for a remaining minimum system information (RMSI) control resource set (CORESET) (configured by physical broadcast channel (PBCH)). For example, in an uplink case, the initial BWP may be given by system information block (SIB) for a random access procedure. For example, the default BWP may be configured by a higher layer. For example, an initial value of the default BWP may be an initial DL BWP. For energy saving, if the UE fails to detect downlink control information (DCI) during a specific period, the UE may switch the active BWP of the UE to the default BWP.

Meanwhile, the BWP may be defined for SL. The same SL BWP may be used in transmission and reception. For example, a transmitting UE may transmit an SL channel or an SL signal on a specific BWP, and a receiving UE may receive the SL channel or the SL signal on the specific BWP. In a licensed carrier, the SL BWP may be defined separately from a Uu BWP, and the SL BWP may have configuration signaling separate from the Uu BWP. For example, the UE may receive a configuration for the SL BWP from the BS/network. For example, the UE may receive a configuration for the Uu BWP from the BS/network. The SL BWP may be (pre-)configured in a carrier with respect to an out-of-coverage NR V2X UE and an RRC_IDLE UE. For the UE in the RRC_CONNECTED mode, at least one SL BWP may be activated in the carrier.

FIG. 5 shows an example of a BWP, based on an embodiment of the present disclosure. The embodiment of FIG. 5 may be combined with various embodiments of the present disclosure. It is assumed in the embodiment of FIG. 5 that the number of BWPs is 3.

Referring to FIG. 5, a common resource block (CRB) may be a carrier resource block numbered from one end of a carrier band to the other end thereof. In addition, the PRB may be a resource block numbered within each BWP. A point A may indicate a common reference point for a resource block grid.

The BWP may be configured by a point A, an offset N_{start}^{BWP} from the point A, and a bandwidth N_{size}^{BWP} . For example, the point A may be an external reference point of a PRB of a carrier in which a subcarrier 0 of all numerologies (e.g., all numerologies supported by a network on that carrier) is aligned. For example, the offset may be a PRB interval between a lowest subcarrier and the point A in a given numerology. For example, the bandwidth may be the number of PRBs in the given numerology.

Hereinafter, V2X or SL communication will be described.

A sidelink synchronization signal (SLSS) may include a primary sidelink synchronization signal (PSSS) and a secondary sidelink synchronization signal (SSSS), as an SL-specific sequence. The PSSS may be referred to as a sidelink primary synchronization signal (S-PSS), and the SSSS may be referred to as a sidelink secondary synchronization signal (S-SSS). For example, length-127 M-sequences may be used for the S-PSS, and length-127 gold sequences may be used for the S-SSS. For example, a UE may use the S-PSS for initial signal detection and for synchronization acquisition. For example, the UE may use the S-PSS and the S-SSS for acquisition of detailed synchronization and for detection of a synchronization signal ID.

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A physical sidelink broadcast channel (PSBCH) may be a (broadcast) channel for transmitting default (system) information which must be first known by the UE before SL signal transmission/reception. For example, the default information may be information related to SLSS, a duplex mode (DM), a time division duplex (TDD) uplink/downlink (UL/DL) configuration, information related to a resource pool, a type of an application related to the SLSS, a subframe offset, broadcast information, or the like. For example, for evaluation of PSBCH performance, in NR V2X, a payload size of the PSBCH may be 56 bits including 24-bit cyclic redundancy check (CRC).

The S-PSS, the S-SSS, and the PSBCH may be included in a block format (e.g., SL synchronization signal (SS)/PSBCH block, hereinafter, sidelink-synchronization signal block (S-SSB)) supporting periodical transmission. The S-SSB may have the same numerology (i.e., SCS and CP length) as a physical sidelink control channel (PSCCH)/physical sidelink shared channel (PSSCH) in a carrier, and a transmission bandwidth may exist within a (pre-)configured sidelink (SL) BWP. For example, the S-SSB may have a bandwidth of 11 resource blocks (RBs). For example, the PSBCH may exist across 11 RBs. In addition, a frequency position of the S-SSB may be (pre-)configured. Accordingly, the UE does not have to perform hypothesis detection at frequency to discover the S-SSB in the carrier.

FIG. 6 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, based on an embodiment of the present disclosure. The embodiment of FIG. 6 may be combined with various embodiments of the present disclosure. In various embodiments of the present disclosure, the transmission mode may be called a mode or a resource allocation mode. Hereinafter, for convenience of explanation, in LTE, the transmission mode may be called an LTE transmission mode. In NR, the transmission mode may be called an NR resource allocation mode.

For example, (a) of FIG. 6 shows a UE operation related to an LTE transmission mode 1 or an LTE transmission mode 3. Alternatively, for example, (a) of FIG. 6 shows a UE operation related to an NR resource allocation mode 1. For example, the LTE transmission mode 1 may be applied to general SL communication, and the LTE transmission mode 3 may be applied to V2X communication.

For example, (b) of FIG. 6 shows a UE operation related to an LTE transmission mode 2 or an LTE transmission mode 4. Alternatively, for example, (b) of FIG. 6 shows a UE operation related to an NR resource allocation mode 2.

Referring to (a) of FIG. 6, in the LTE transmission mode 1, the LTE transmission mode 3, or the NR resource allocation mode 1, a base station may schedule SL resource(s) to be used by a UE for SL transmission. For example, in step S600, a base station may transmit information related to SL resource(s) and/or information related to UL resource(s) to a first UE. For example, the UL resource(s) may include PUCCH resource(s) and/or PUSCH resource(s). For example, the UL resource(s) may be resource(s) for reporting SL HARQ feedback to the base station.

For example, the first UE may receive information related to dynamic grant (DG) resource(s) and/or information related to configured grant (CG) resource(s) from the base station. For example, the CG resource(s) may include CG type 1 resource(s) or CG type 2 resource(s). In the present disclosure, the DG resource(s) may be resource(s) configured/allocated by the base station to the first UE through a downlink control information (DCI). In the present disclosure, the CG resource(s) may be (periodic) resource(s)

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configured/allocated by the base station to the first UE through a DCI and/or an RRC message. For example, in the case of the CG type 1 resource(s), the base station may transmit an RRC message including information related to CG resource(s) to the first UE. For example, in the case of the CG type 2 resource(s), the base station may transmit an RRC message including information related to CG resource(s) to the first UE, and the base station may transmit a DCI related to activation or release of the CG resource(s) to the first UE.

In step S610, the first UE may transmit a PSCCH (e.g., sidelink control information (SCI) or 1st-stage SCI) to a second UE based on the resource scheduling. In step S620, the first UE may transmit a PSSCH (e.g., 2nd-stage SCI, MAC PDU, data, etc.) related to the PSCCH to the second UE. In step S630, the first UE may receive a PSFCH related to the PSCCH/PSSCH from the second UE. For example, HARQ feedback information (e.g., NACK information or ACK information) may be received from the second UE through the PSFCH. In step S640, the first UE may transmit/report HARQ feedback information to the base station through the PUCCH or the PUSCH. For example, the HARQ feedback information reported to the base station may be information generated by the first UE based on the HARQ feedback information received from the second UE. For example, the HARQ feedback information reported to the base station may be information generated by the first UE based on a pre-configured rule. For example, the DCI may be a DCI for SL scheduling. For example, a format of the DCI may be a DCI format 3_0 or a DCI format 3_1.

Referring to (b) of FIG. 6, in the LTE transmission mode 2, the LTE transmission mode 4, or the NR resource allocation mode 2, a UE may determine SL transmission resource(s) within SL resource(s) configured by a base station/network or pre-configured SL resource(s). For example, the configured SL resource(s) or the pre-configured SL resource(s) may be a resource pool. For example, the UE may autonomously select or schedule resource(s) for SL transmission. For example, the UE may perform SL communication by autonomously selecting resource(s) within the configured resource pool. For example, the UE may autonomously select resource(s) within a selection window by performing a sensing procedure and a resource (re)selection procedure. For example, the sensing may be performed in a unit of subchannel(s). For example, in step S610, a first UE which has selected resource(s) from a resource pool by itself may transmit a PSCCH (e.g., sidelink control information (SCI) or 1st-stage SCI) to a second UE by using the resource(s). In step S620, the first UE may transmit a PSSCH (e.g., 2nd-stage SCI, MAC PDU, data, etc.) related to the PSCCH to the second UE. In step S630, the first UE may receive a PSFCH related to the PSCCH/PSSCH from the second UE.

Referring to (a) or (b) of FIG. 6, for example, the first UE may transmit a SCI to the second UE through the PSCCH. Alternatively, for example, the first UE may transmit two consecutive SCIs (e.g., 2-stage SCI) to the second UE through the PSCCH and/or the PSSCH. In this case, the second UE may decode two consecutive SCIs (e.g., 2-stage SCI) to receive the PSSCH from the first UE. In the present disclosure, a SCI transmitted through a PSCCH may be referred to as a 1st SCI, a first SCI, a 1st-stage SCI or a 1st-stage SCI format, and a SCI transmitted through a PSSCH may be referred to as a 2nd SCI, a second SCI, a 2nd-stage SCI or a 2nd-stage SCI format. For example, the

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1st-stage SCI format may include a SCI format 1-A, and the 2nd-stage SCI format may include a SCI format 2-A and/or a SCI format 2-B.

Referring to (a) or (b) of FIG. 6, in step S630, the first UE may receive the PSFCH. For example, the first UE and the second UE may determine a PSFCH resource, and the second UE may transmit HARQ feedback to the first UE using the PSFCH resource.

Referring to (a) of FIG. 6, in step S640, the first UE may transmit SL HARQ feedback to the base station through the PUCCH and/or the PUSCH.

FIG. 7 shows three cast types, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 7 may be combined with various embodiments of the present disclosure. Specifically, FIG. 7(a) shows broadcast-type SL communication, FIG. 7(b) shows unicast type-SL communication, and FIG. 7(c) shows groupcast-type SL communication. In case of the unicast-type SL communication, a UE may perform one-to-one communication with respect to another UE. In case of the groupcast-type SL transmission, the UE may perform SL communication with respect to one or more UEs in a group to which the UE belongs. In various embodiments of the present disclosure, SL groupcast communication may be replaced with SL multicast communication, SL one-to-many communication, or the like.

In this specification, the “configure or define” wording may be interpreted as being (pre)configured (via pre-defined signaling (e.g., SIB, MAC signaling, RRC signaling)) from a base station or a network. For example, “A may be configured” may include “that a base station or network (pre-)configures/defines or informs A for a UE”. Alternatively, the wording “configure or define” may be interpreted as being configured or defined in advance by a system. For example, “A may be configured” may include “A is configured/defined in advance by a system”.

FIG. 8 shows an example of an architecture in a 5G system in which positioning for a UE connected to a Next Generation-Radio Access Network (NG-RAN) or E-UTRAN is possible, according to an embodiment of the present disclosure. The embodiment of FIG. 8 may be combined with various embodiments of the present disclosure.

Referring to FIG. 8, an AMF may receive a request for a location service related to a specific target UE from a different entity such as a gateway mobile location center (GMLC), or may determine to start the location service in the AMF itself instead of the specific target UE. Then, the AMF may transmit a location service request to a location management function (LMF). Upon receiving the location service request, the LMF may process the location service request and return a processing request including an estimated position or the like of the UE to the AMF. Meanwhile, if the location service request is received from the different entity such as GMLC other than the AMF, the AMF may transfer to the different entity the processing request received from the LMF.

A new generation evolved-NB (ng-eNB) and a gNB are network elements of NG-RAN capable of providing a measurement result for position estimation, and may measure a radio signal for a target UE and may transfer a resultant value to the LMF. In addition, the ng-eNB may control several transmission points (TPs) such as remote radio heads or PRS-dedicated TPs supporting a positioning reference signal (PRS)-based beacon system for E-UTRA.

The LMF may be connected to an enhanced serving mobile location center (E-SMLC), and the E-SMLC may

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allow the LMF to access E-UTRAN. For example, the E-SMLC may allow the LMF to support observed time difference of arrival (OTDOA), which is one of positioning methods of E-UTRAN, by using downlink measurement obtained by a target UE through a signal transmitted from the gNB and/or the PRS-dedicated TPs in the E-UTRAN.

Meanwhile, the LMF may be connected to an SUPL location platform (SLP). The LMF may support and manage different location determining services for respective target UEs. The LMF may interact with a serving ng-eNB or serving gNB for the target UE to obtain location measurement of the UE. For positioning of the target UE, the LMF may determine a positioning method based on a location service (LCS) client type, a requested quality of service (QoS), UE positioning capabilities, gNB positioning capabilities, and ng-eNB positioning capabilities, or the like, and may apply such a positioning method to the serving gNB and/or the serving ng-eNB. In addition, the LMF may determine additional information such as a position estimation value for the target UE and accuracy of position estimation and speed. The SLP is a secure user plane location (SUPL) entity in charge of positioning through a user plane.

The UE may measure a downlink signal through NG-RAN, E-UTRAN, and/or other sources such as different global navigation satellite system (GNSS) and terrestrial beacon system (TBS), wireless local access network (WLAN) access points, Bluetooth beacons, UE barometric pressure sensors or the like. The UE may include an LCS application. The UE may communicate with a network to which the UE has access, or may access the LCS application through another application included in the UE. The LCS application may include a measurement and calculation function required to determine a position of the UE. For example, the UE may include an independent positioning function such as a global positioning system (GPS), and may report the position of the UE independent of NG-RAN transmission. Positioning information obtained independently as such may be utilized as assistance information of the positioning information obtained from the network.

FIG. 9 shows an implementation example of a network for measuring a position of a UE, according to an embodiment of the present disclosure. The embodiment of FIG. 9 may be combined with various embodiments of the present disclosure.

When the UE is in a connection management (CM)-IDLE state, if an AMF receives a location service request, the AMF may establish a signaling connection with the UE, and may request for a network trigger service to allocate a specific serving gNB or ng-eNB. Such an operational process is omitted in FIG. 9. That is, it may be assumed in FIG. 9 that the UE is in a connected mode. However, due to signaling and data inactivation or the like, the signaling connection may be released by NG-RAN while a positioning process is performed.

A network operation process for measuring a position of a UE will be described in detail with reference to FIG. 9. In step S910, a 5GC entity such as GMLC may request a serving AMF to provide a location service for measuring a position of a target UE. However, even if the GMLC does not request for the location service, based on step S915, the serving AMF may determine that the location service for measuring the position of the target UE is required. For example, to measure the position of the UE for an emergency call, the serving AMF may determine to directly perform the location service.

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Thereafter, the AMF may transmit the location service request to an LMF based on step S920, and the LMF may start location procedures to obtain position measurement data or position measurement assistance data together with a serving ng-eNB and a serving gNB. Additionally, based on step S935, the LMF may start location procedures for downlink positioning together with the UE. For example, the LMF may transmit assistance data defined in 3GPP TS 36.355, or may obtain a position estimation value or a position measurement value. Meanwhile, step S935 may be performed additionally after step S930 is performed, or may be performed instead of step S930.

In step S940, the LMF may provide a location service response to the AMF. In addition, the location service response may include information regarding whether position estimation of the UE is successful and a position estimation value of the UE. Thereafter, if the procedure of FIG. 9 is initiated by step S910, in step S950, the AMF may transfer the location service response to a 5GC entity such as GMLC, and if the procedure of FIG. 9 is initiated by step S915, in step S955, the AMF may use the location service response to provide a location service related to an emergency call or the like.

FIG. 10 shows an example of a protocol layer used to support LTE Positioning Protocol (LPP) message transmission between an LMF and a UE, according to an embodiment of the present disclosure. The embodiment of FIG. 10 may be combined with various embodiments of the present disclosure.

An LPP PDU may be transmitted through a NAS PDU between an AMF and the UE. Referring to FIG. 10, an LPP may be terminated between a target device (e.g., a UE in a control plane or an SUPL enabled terminal (SET) in a user plane) and a location server (e.g., an LMF in the control plane and an SLP in the user plane). The LPP message may be transferred in a form of a transparent PDU through an intermediary network interface by using a proper protocol such as an NG application protocol (NGAP) through an NG-control plane (NG-C) interface and NAS/RRC or the like through an NR-Uu interface. The LPP protocol may enable positioning for NR and LTE by using various positioning methods.

For example, based on the LPP protocol, the target device and the location server may exchange mutual capability information, assistance data for positioning, and/or location information. In addition, an LPP message may be used to indicate exchange of error information and/or interruption of the LPP procedure.

FIG. 11 shows an example of a protocol layer used to support NR Positioning Protocol A (NRPPa) PDU transmission between an LMF and an NG-RAN node, according to an embodiment of the present disclosure. The embodiment of FIG. 11 may be combined with various embodiments of the present disclosure.

Referring to FIG. 11, the NRPPa may be used for information exchange between the NG-RAN node and the LMF. Specifically, the NRPPa may exchange an enhanced-cell ID (E-CID) for measurement, data for supporting an OTDOA positioning method, and a cell-ID, cell location ID, or the like for an NR cell ID positioning method, transmitted from the ng-eNB to the LMF. Even if there is no information regarding an associated NRPPa transaction, the AMF may route NRPPa PDUs based on a routing ID of an associated LMR through an NG-C interface.

A procedure of an NRPPa protocol for location and data collection may be classified into two types. A first type is a UE associated procedure for transferring information

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regarding a specific UE (e.g., position measurement information or the like), and a second type is a non UE associated procedure for transferring information (e.g., gNB/ng-eNB/TP timing information, etc.) applicable to an NG-RAN node and associated TPs. The two types of the procedure may be independently supported or may be simultaneously supported.

Meanwhile, examples of positioning methods supported in NG-RAN may include GNSS, OTDOA, enhanced cell ID (E-CID), barometric pressure sensor positioning, WLAN positioning, Bluetooth positioning and terrestrial beacon system (TBS), uplink time difference of arrival (UTDOA), etc.

(1) OTDOA (Observed Time Difference of Arrival)

FIG. 12 shows an Observed Time Difference Of Arrival (OTDOA) positioning method according to an embodiment of the present disclosure. The embodiment of FIG. 12 may be combined with various embodiments of the present disclosure.

Referring to FIG. 12, the OTDOA positioning method uses measurement timing of downlink signals received by a UE from an eNB, an ng-eNB, and a plurality of TPs including a PRS-dedicated TP. The UE measures timing of downlink signals received by using location assistance data received from a location server. In addition, a position of the UE may be determined based on such a measurement result and geometric coordinates of neighboring TPs.

A UE connected to a gNB may request for a measurement gap for OTDOA measurement from the TP. If the UE cannot recognize a single frequency network (SFN) for at least one TP in the OTDOA assistance data, the UE may use an autonomous gap to obtain an SNF of an OTDOA reference cell before the measurement gap is requested to perform reference signal time difference (RSTD) measurement.

Herein, the RSTD may be defined based on a smallest relative time difference between boundaries of two subframes received respectively from a reference cell and a measurement cell. That is, the RSTD may be calculated based on a relative time difference between a start time of a subframe received from the measurement cell and a start time of a subframe of a reference cell closest to the start time of the subframe received from the measurement cell. Meanwhile, the reference cell may be selected by the UE.

For correct OTDOA measurement, it may be necessary to measure a time of arrival (TOA) of a signal received from three or more TPs or BSs geometrically distributed. For example, a TOA may be measured for each of a TP1, a TP2, and a TP3, and RSTD for TP 1-TP 2, RSTD for TP 2-TP 3, and RSTD for TP 3-TP 1 may be calculated for the three TOAs. Based on this, a geometric hyperbola may be determined, and a point at which these hyperbolas intersect may be estimated as a position of a UE. In this case, since accuracy and/or uncertainty for each TOA measurement may be present, the estimated position of the UE may be known as a specific range based on measurement uncertainty.

For example, RSTD for two TPs may be calculated based on Equation 1.

$$RSTD_{i,1} = \frac{\sqrt{(x_t - x_1)^2 + (y_t - y_1)^2}}{c} - \frac{\sqrt{(x_t - x_1)^2 + (y_t - y_1)^2}}{c} + (T_i - T_1) + (n_i - n_1) \quad [\text{Equation 1}]$$

Herein, c may be the speed of light, $\{x_t, y_t\}$ may be a (unknown) coordinate of a target UE, $\{x_i, y_i\}$ may be a coordinate of a (known) TP, and $\{x_1, y_1\}$ may be a coordinate of a reference TP (or another TP). Herein, $(T_i - T_1)$ may be referred to as “real time differences (RTDs)” as a transmission time offset between two TPs, and n_i, n_1 may represent values related to UE TOA measurement errors.

(2) E-CID (Enhanced Cell ID)

In a cell ID (CID) positioning method, a position of a UE may be measured through geometric information of a serving ng-eNB, serving gNB, and/or serving cell of the UE. For example, the geometric information of the serving ng-eNB, serving gNB, and/or serving cell may be obtained through paging, registration, or the like.

Meanwhile, in addition to the CID positioning method, an E-CID positioning method may use additional UE measurement and/or NG-RAN radio resources or the like to improve a UE position estimation value. In the E-CID positioning method, although some of the measurement methods which are the same as those used in a measurement control system of an RRC protocol may be used, additional measurement is not performed in general only for position measurement of the UE. In other words, a measurement configuration or a measurement control message may not be provided additionally to measure the position of the UE. Also, the UE may not expect that an additional measurement operation only for position measurement will be requested, and may report a measurement value obtained through measurement methods in which the UE can perform measurement in a general manner.

For example, the serving gNB may use an E-UTRA measurement value provided from the UE to implement the E-CID positioning method.

Examples of a measurement element that can be used for E-CID positioning may be as follows.

UE measurement: E-UTRA reference signal received power (RSRP), E-UTRA reference signal received quality (RSRQ), UE E-UTRA Rx-Tx Time difference, GSM EDGE random access network (GERAN)/

WLAN reference signal strength indication (RSSI), UTRAN common pilot channel (CPICH) received signal code power (RSCP), UTRAN CPICH Ec/Io

E-UTRAN measurement: ng-eNB Rx-Tx Time difference, timing advance (TADV), angle of arrival (AoA)

Herein, the TADV may be classified into Type 1 and Type 2 as follows.

TADV Type 1 = (ng-eNB Rx-Tx time difference) + (UE E-UTRA Rx-Tx time difference)

TADV Type 2 = ng-eNB Rx-Tx time difference

Meanwhile, AoA may be used to measure a direction of the UE. The AoA may be defined as an estimation angle with respect to the position of the UE counterclockwise from a BS/TP. In this case, a geographic reference direction may be north. The BS/TP may use an uplink signal such as a sounding reference signal (SRS) and/or a demodulation reference signal (DMRS) for AoA measurement. In addition, the larger the arrangement of the antenna array, the higher the measurement accuracy of the AoA. When the antenna arrays are arranged with the same interval, signals received from adjacent antenna elements may have a constant phase-rotate.

(3) UTDOA (Uplink Time Difference of Arrival)

UTDOA is a method of determining a position of a UE by estimating an arrival time of SRS. When calculating an estimated SRS arrival time, the position of the UE may be estimated through an arrival time difference with respect to another cell (or BS/TP) by using a serving cell as a reference cell. In order to implement the UTDOA, E-SMLC may indicate a serving cell of a target UE to indicate SRS transmission to the target UE. In addition, the E-SMLC may provide a configuration such as whether the SRS is periodic/apperiodical, a bandwidth, frequency/group/sequence hopping, or the like.

Referring to the standard document, some procedures and technical specifications related to the present disclosure are as follows.

TABLE 5

Reference signal time difference (RSTD) for E-UTRA	
Definition	The relative timing difference between the E-UTRA neighbour cell j and the E-UTRA reference cell i , defined as $T_{SubframeRxj} - T_{SubframeRxi}$ where: $T_{SubframeRxj}$ is the time when the UE receives the start of one subframe from E-UTRA cell j $T_{SubframeRxi}$ is the time when the UE receives the corresponding start of one subframe from E-UTRA cell i that is closest in time to the subframe received from E-UTRA cell j . The reference point for the observed subframe time difference shall be the antenna connector of the UE.
Applicable for	RRC_CONNECTED inter-RAT

TABLE 6

DL PRS reference signal received power (DL PRS-RSRP)	
Definition	DL PRS reference signal received power (DL PRS-RSRP), is defined as the linear average over the power contributions (in [W]) of the resource elements that carry DL PRS reference signals configured for RSRP measurements within the considered measurement frequency bandwidth. For frequency range 1, the reference point for the DL PRS-RSRP shall be the antenna connector of the UE. For frequency range 2, DL PRS-RSRP shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch. For frequency range 1 and 2, if receiver diversity is in use by the UE, the reported DL PRS-RSRP value shall not be lower than the corresponding DL PRS-RSRP of any of the individual receiver branches.

TABLE 6-continued

DL PRS reference signal received power (DL PRS-RSRP)	
Applicable for	RRC_CONNECTED intra-frequency, RRC_CONNECTED inter-frequency

TABLE 7

DL relative signal time difference (DL RSTD)	
Definition	DL relative timing difference (DL RSTD) between the positioning node j and the reference positioning node i, is defined as $T_{SubframeRxj} - T_{SubframeRxi}$ Where: $T_{SubframeRxj}$ is the time when the UE receives the start of one subframe from positioning node j. $T_{SubframeRxi}$ is the time when the UE receives the corresponding start of one subframe from positioning node i that is closest in time to the subframe received from positioning node j. Multiple DL PRS resources can be used to determine the start of one subframe from a positioning node. For frequency range 1, the reference point for the DL RSTD shall be the antenna connector of the UE. For frequency range 2, the reference point for the DL RSTD shall be the antenna of the UE.
Applicable for	RRC_CONNECTED intra-frequency RRC_CONNECTED inter-frequency

TABLE 8

UE Rx – Tx time difference	
Definition	The UE Rx – Tx time difference is defined as $T_{UE-RX} - T_{UE-TX}$ Where: T_{UE-RX} is the UE received timing of downlink subframe #i from a positioning node, defined by the first detected path in time. T_{UE-TX} is the UE transmit timing of uplink subframe #j that is closest in time to the subframe #i received from the positioning node. Multiple DL PRS resources can be used to determine the start of one subframe of the first arrival path of the positioning node. For frequency range 1, the reference point for T_{UE-RX} measurement shall be the Rx antenna connector of the UE and the reference point for T_{UE-TX} measurement shall be the Tx antenna connector of the UE. For frequency range 2, the reference point for T_{UE-RX} measurement shall be the Rx antenna of the UE and the reference point for T_{UE-TX} measurement shall be the Tx antenna of the UE.
Applicable for	RRC_CONNECTED intra-frequency RRC_CONNECTED inter-frequency

TABLE 9

UL Relative Time of Arrival (TUL-RTOA)	
Definition	[The UL Relative Time of Arrival ($T_{UL-RTOA}$) is the beginning of subframe i containing SRS received in positioning node j, relative to the configurable reference time.] Multiple SRS resources for positioning can be used to determine the beginning of one subframe containing SRS received at a positioning node. The reference point for $T_{UL-RTOA}$ shall be: - for type 1-C base station TS 38.104 [9]: the Rx antenna connector, - for type 1-O or 2-O base station TS 38.104 [9]: the Rx antenna, - for type 1-H base station TS 38.104 [9]: the Rx Transceiver Array Boundary connector.

TABLE 10

gNB Rx – Tx time difference	
Definition	The gNB Rx – Tx time difference is defined as $T_{gNB-RX} - T_{gNB-TX}$ Where: T_{gNB-RX} is the positioning node received timing of uplink subframe #i containing SRS associated with UE, defined by the first detected path in time. T_{gNB-TX} is the positioning node transmit timing of downlink subframe #j that is closest in time to the subframe #i received from the UE. Multiple SRS resources for positioning can be used to determine the start of one subframe containing SRS. The reference point for T_{gNB-RX} shall be: - for type 1-C base station TS 38.104 [9]: the Rx antenna connector, - for type 1-O or 2-O base station TS 38.104 [9]: the Rx antenna, - for type 1-H base station TS 38.104 [9]: the Rx Transceiver Array Boundary connector. The reference point for T_{gNB-TX} shall be: - for type 1-C base station TS 38.104 [9]: the Tx antenna connector, - for type 1-O or 2-O base station TS 38.104 [9]: the Tx antenna, - for type 1-H base station TS 38.104 [9]: the Tx Transceiver Array Boundary connector.

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TABLE 11

UL Angle of Arrival (UL AoA)	
Definition	UL Angle of Arrival (UL AoA) is defined as the estimated azimuth angle and vertical angle of a UE with respect to a reference direction, wherein the reference direction is defined: - In the global coordinate system (GCS), wherein estimated azimuth angle is measured relative to geographical North and is positive in a counter-clockwise direction and estimated vertical angle is measured relative to zenith and positive to horizontal direction - In the local coordinate system (LCS), wherein estimated azimuth angle is measured relative to x-axis of LCS and positive in a counter-clockwise direction and estimated vertical angle is measured relative to z-axis of LCS and positive to x-y plane direction. The bearing, downtilt and slant angles of LCS are defined according to TS 38.901 [14]. The UL AoA is determined at the gNB antenna for an UL channel corresponding to this UE.

TABLE 12

UL SRS reference signal received power (UL SRS-RSRP)	
Definition	UL SRS reference signal received power (UL SRS-RSRP) is defined as linear average of the power contributions (in [W]) of the resource elements carrying sounding reference signals (SRS). UL SRS-RSRP shall be measured over the configured resource elements within the considered measurement frequency bandwidth in the configured measurement time occasions. For frequency range 1, the reference point for the UL SRS-RSRP shall be the antenna connector of the gNB. For frequency range 2, UL SRS-RSRP shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch. For frequency range 1 and 2, if receiver diversity is in use by the gNB, the reported UL SRS-RSRP value shall not be lower than the corresponding UL SRS-RSRP of any of the individual receiver branches.

TABLE 13

14.1.1.6	UE procedure for determining the subset of resources to be reported to higher layers in PSSCH resource selection in sidelink transmission mode 4 and in sensing measurement in sidelink transmission mode 3 In sidelink transmission mode 4, when requested by higher layers in subframe n for a carrier, the UE shall determine the set of resources to be reported to higher layers for PSSCH transmission according to the steps described in this Subclause. Parameters L_{subCH} the number of sub-channels to be used for the PSSCH transmission in a subframe, P_{rsvp_TX} the resource reservation interval, and $prio_{TX}$ the priority to be transmitted in the associated SCI format 1 by the UE are all provided by higher layers (described in [8]). C_{resel} is determined according to Subclause 14.1.1.4B.
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TABLE 13-continued

In sidelink transmission mode 3, when requested by higher layers in subframe n for a carrier, the UE shall determine the set of resources to be reported to higher layers in sensing measurement according to the steps described in this Subclause. Parameters L_{subCH} , P_{rsrp_TX} and $prio_{TX}$ are all provided by higher layers (described in [11]). C_{resel} is determined by $C_{resel} = 10 \times SL_RESOURCE_RESELECTION_COUNTER$, where $SL_RESOURCE_RESELECTION_COUNTER$ is provided by higher layers [11].

...

If partial sensing is configured by higher layers then the following steps are used:

- 1) A candidate single-subframe resource for PSSCH transmission $R_{x,y}$ is defined as a set of L_{subCH} contiguous sub-channels with sub-channel $x+j$ in subframe t_y^{SL} where $j=0, \dots, L_{subCH}-1$. The UE shall determine by its implementation a set of subframes which consists of at least Y subframes within the time interval $[n + T_1, n + T_2]$ where selections of T_1 and T_2 are up to UE implementations under $T_1 \leq 4$ and $T_{2min}(prio_{TX}) \leq T_2 \leq 100$, if $T_{2min}(prio_{TX})$ is provided by higher layers for $prio_{TX}$, otherwise $20 \leq T_2 \leq 100$. UE selection of T_2 shall fulfil the latency requirement and Y shall be greater than or equal to the high layer parameter $minNumCandidateSF$. The UE shall assume that any set of L_{subCH} contiguous sub-channels included in the corresponding PSSCH resource pool (described in 14.1.5) within the determined set of subframes correspond to one candidate single-subframe resource. The total number of the candidate single-subframe resources is denoted by M_{total} .
- 2) If a subframe t_y^{SL} is included in the set of subframes in Step 1, the UE shall monitor any subframe $t_{y-k \times P_{step}}^{SL}$ if k -th bit of the high layer parameter $gapCandidateSensing$ is set to 1. The UE shall perform the behaviour in the following steps based on PSSCH decoded and S-RSSI measured in these subframes.
- 3) The parameter $Th_{a,b}$ is set to the value indicated by the i -th SL-ThresPSSCH-RSRP field in SL-ThresPSSCH-RSRP-List where $i = (a - 1) \times 8 + b$.

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TABLE 14

- 4) The set S_A is initialized to the union of all the candidate single-subframe resources. The set S_B is initialized to an empty set.
- 5) The UE shall exclude any candidate single-subframe resource $R_{x,y}$ from the set S_A if it meets all the following conditions:
the UE receives an SCI format 1 in subframe t_m^{SL} , and "Resource reservation" field and "Priority" field in the received SCI format 1 indicate the values P_{rsrp_RX} and $prio_{RX}$, respectively according to Subclause 14.2.1.
PSSCH-RSRP measurement according to the received SCI format 1 is higher than $Th_{prio_{TX}, prio_{RX}}$
the SCI format received in subframe t_m^{SL} or the same SCI format 1 which is assumed to be received in subframe(s) $t_{m+q \times P_{step} \times P_{rsrp_RX}}^{SL}$ determines according to 14.1.1.4C the set of resource blocks and subframes which overlaps with $R_{x,y+j \times P_{rsrp_TX}}$ for $q = 1, 2, \dots, Q$ and
$$j = 0, 1, \dots, C_{resel} - 1$$
. Here $Q = \frac{1}{P_{rsrp_RX}}$ if $P_{rsrp_RX} < 1$ and
 $y' - m \leq P_{step} \times P_{rsrp_RX} + P_{step}$, where $t_{y'}^{SL}$ is the last subframe of the Y subframes, and $Q = 1$ otherwise.
- 6) If the number of candidate single-subframe resources remaining in the set S_A is smaller than $0.2 \times M_{total}$, then Step 4 is repeated with $Th_{a,b}$ increased by 3 dB.
- 7) For a candidate single-subframe resource $R_{x,y}$ remaining in the set S_A , the metric $E_{x,y}$ is defined as the linear average of S-RSSI measured in sub-channels $x+k$ for $k = 0, \dots, L_{subCH} - 1$ in the monitored subframes in Step 2 that can be expressed by $t_{y-P_{step} \times j}^{SL}$ for a non-negative integer j .
- 8) The UE moves the candidate single-subframe resource $R_{x,y}$ with the smallest metric $E_{x,y}$ from the set S_A to S_B . This step is repeated until the number of candidate single-subframe resources in the set S_B becomes greater than or equal to $0.2 \times M_{total}$.
- 9) When the UE is configured by upper layers to transmit using resource pools on multiple carriers, it shall exclude a candidate single-subframe resource $R_{x,y}$ from S_B if the UE does not support transmission in the candidate single-subframe resource in the carrier under the assumption that transmissions take place in other carrier(s) using the already selected resources due to its limitation in the number of simultaneous transmission carriers, its limitation in the supported carrier combinations, or interruption for RF retuning time [10].

TABLE 15

The UE shall report set S_B to higher layers.

If transmission based on random selection is configured by upper layers and when the UE is configured by upper layers to transmit using resource pools on multiple carriers, the following steps are used:

- 1) A candidate single-subframe resource for PSSCH transmission $R_{x,y}$ is defined as a set of L_{subCH} contiguous sub-channels with sub-channel $x+j$ in subframe t_y^{SL} where

TABLE 15-continued

	$j = 0, \dots, L_{subCH} - 1$. The UE shall assume that any set of L_{subCH} contiguous sub-channels included in the corresponding PSSCH resource pool (described in 14.1.5) within the time interval $[n+T_1, n+T_2]$ corresponds to one candidate single-subframe resource, where selections of T_1 and T_2 are up to UE implementations under $T_1 \leq 4$ and $T_{2min}(\text{prio}_{TX}) \leq T_2 \leq 100$, if $T_{2min}(\text{prio}_{TX})$ is provided by higher layers for prio_{TX} , otherwise $20 \leq T_2 \leq 10$. UE selection of T_2 shall fulfil the latency requirement. The total number of the candidate single-subframe resources is denoted by M_{total} .
2)	The set S_A is initialized to the union of all the candidate single-subframe resources. The set S_B is initialized to an empty set.
3)	The UE moves the candidate single-subframe resource $R_{x,j}$ from the set S_A to S_B .
4)	The UE shall exclude a candidate single-subframe resource $R_{x,j}$ from S_B if the UE does not support transmission in the candidate single-subframe resource in the carrier under the assumption that transmissions take place in other carrier(s) using the already selected resources due to its limitation in the number of simultaneous transmission carriers, its limitation in the supported carrier combinations, or interruption for RF retuning time [10].

The UE shall report set S_B to higher layers.

TABLE 16

UE procedure for determining the subset of resources to be reported to higher layers in PSSCH resource selection in sidelink resource allocation mode 2

In resource allocation mode 2, the higher layer can request the UE to determine a subset of resources from which the higher layer will select resources for PSSCH/PSCCH transmission. To trigger this procedure, in slot n , the higher layer provides the following parameters for this PSSCH/PSCCH transmission:

- the resource pool from which the resources are to be reported;
- L1 priority, prio_{TX} ;
- the remaining packet delay budget;
- the number of sub-channels to be used for the PSSCH/PSCCH transmission in a slot, L_{subCH} ;
- optionally, the resource reservation interval, P_{rsvp_TX} , in units of msec.
- if the higher layer requests the UE to determine a subset of resources from which the higher layer will select resources for PSSCH/PSCCH transmission as part of re-evaluation or pre-emption procedure, the higher layer provides a set of resources $(r_0, r_1, r_2, \dots)$ which may be subject to re-evaluation and a set of resources $(r'_0, r'_1, r'_2, \dots)$ which may be subject to pre-emption.
- it is up to UE implementation to determine the subset of resources as requested by higher layers before or after the slot $r'_i - T_3$, where r'_i is the slot with the smallest slot index among $(r_0, r_1, r_2, \dots)$ and $(r'_0, r'_1, r'_2, \dots)$, and T_3 is equal to $T_{proc,1}^{SL}$, where $T_{proc,1}^{SL}$ is defined in slots in Table 8.1.4-2 where μ_{SL} is the SCS configuration of the SL BWP

The following higher layer parameters affect this procedure:

- sl-SelectionWindowList: internal parameter T_{2min} is set to the corresponding value from higher layer parameter sl-SelectionWindowList for the given value of prio_{TX} .
- sl-Thres-RSRP-List: this higher layer parameter provides an RSRP threshold for each combination (p_i, p_j) , where p_i is the value of the priority field in a received SCI format 1-A and p_j is the priority of the transmission of the UE selecting resources; for a given invocation of this procedure, $p_j = \text{prio}_{TX}$.
- sl-RS-ForSensing selects if the UE uses the PSSCH-RSRP or PSCCH-RSRP measurement, as defined in clause 8.4.2.1.
- sl-ResourceReservePeriodList
- sl-SensingWindow: internal parameter T_0 is defined as the number of slots corresponding to sl-SensingWindow msec
- sl-TxPercentageList: internal parameter X for a given prio_{TX} is defined as sl-TxPercentageList (prio_{TX}) converted from percentage to ratio
- sl-PreemptionEnable: if sl-PreemptionEnable is provided, and if it is not equal to 'enabled', internal parameter prio_{pre} is set to the higher layer provided parameter sl-PreemptionEnable

The resource reservation interval, P_{rsvp_TX} , if provided, is converted from units of msec to units of logical slots, resulting in P'_{rsvp_TX} according to clause 8.1.7.

Notation:

$(t'_0, t'_1, t'_2, \dots)$ denotes the set of slots which belongs to the sidelink resource pool and is defined in Clause 8.

TABLE 17

The following steps are used:

- 1) A candidate single-slot resource for transmission $R_{x,y}$ is defined as a set of L_{subCH} contiguous sub-channels with sub-channel $x + j$ in slot t_y^{SL} where $j = 0, \dots, L_{subCH} - 1$. The UE shall assume that any set of L_{subCH} contiguous sub-channels included in the corresponding resource pool within the time interval $[n + T_1, n + T_2]$ correspond to one candidate single-slot resource, where selection of T_1 is up to UE implementation under $0 \leq T_1 \leq T_{proc,1}^{SL}$, where $T_{proc,1}^{SL}$ is defined in slots in Table 8.1.4-2 where μ_{SL} is the SCS configuration of the SL BWP; if T_{2min} is shorter than the remaining packet delay budget (in slots) then T_2 is up to UE implementation subject to $T_{2min} \leq T_2 \leq$ remaining packet delay budget (in slots); otherwise T_2 is set to the remaining packet delay budget (in slots). The total number of candidate single-slot resources is denoted by M_{total} .
- 2) The sensing window is defined by the range of slots $[n - T_{proc,0}^{SL}, n]$ where T_0 is defined above and $T_{proc,0}^{SL}$ is defined in slots in Table 8.1.4-1 where μ_{SL} is the SCS configuration of the SL BWP. The UE shall monitor slots which belongs to a sidelink resource pool within the sensing window except for those in which its own transmissions occur. The UE shall perform the behaviour in the following steps based on PSSCH decoded and RSRP measured in these slots.
- 3) The internal parameter $Th(p_i, p_j)$ is set to the corresponding value of RSRP threshold indicated by the i -th field in sl-Thres-RSRP-List, where $i = p_i + (p_j - 1) * 8$.
- 4) The set S_A is initialized to the set of all the candidate single-slot resources.
- 5) The UE shall exclude any candidate single-slot resource $R_{x,y}$ from the set S_A if it meets all the following conditions:
the UE has not monitored slot t_m^{SL} in Step 2.
for any periodicity value allowed by the higher layer parameter sl-ResourceReservePeriodList and a hypothetical SCI format 1-A received in slot t_m^{SL} with 'Resource reservation period' field set to that periodicity value and indicating all subchannels of the resource pool in this slot, condition c in step 6 would be met.
- 5a) If the number of candidate single-slot resources $R_{x,y}$ remaining in the set S_A is smaller than $X \cdot M_{total}$, the set S_A is initialized to the set of all the candidate single-slot resources as in step 4.
- 6) The UE shall exclude any candidate single-slot resource $R_{x,y}$ from the set S_A if it meets all the following conditions:
a) the UE receives an SCI format 1-A in slot t_m^{SL} , and 'Resource reservation period' field, if present, and 'Priority' field in the received SCI format 1-A indicate the values P_{rsvp_RX} and $prio_{RX}$, respectively according to Clause 16.4 in [6, TS 38.213];
b) the RSRP measurement performed, according to clause 8.4.2.1 for the received SCI format 1-A, is higher than $Th(prio_{RX}, prio_{TX})$;
c) the SCI format received in slot t_m^{SL} or the same SCI format which, if and only if the 'Resource reservation period' field is present in the received SCI format 1-A, is assumed to be received in slot(s) $t_{m+q:P_{rsvp_RX}}^{SL}$ determines according to clause 8.1.5 the set of resource blocks and slots which overlaps with $R_{x,y+j:P_{rsvp_TX}}$ for $q = 1, 2, \dots, Q$ and $j = 0, 1, \dots, C_{resel} - 1$. Here, P_{rsvp_RS} is P_{rsvp_RX} converted to units of logical slots according to clause 8.1.7,
$$Q = \left\lceil \frac{T_{scal}}{P_{rsvp_RX}} \right\rceil$$

if $P_{rsvp_RX} < T_{scal}$ and $n' - m \leq P_{rsvp_RX}$, where $t_n^{SL} = n$ if slot n belongs to the set $(t_0^{SL}, t_1^{SL}, \dots, t_{T_{max}-1}^{SL})$, otherwise slot t_n^{SL} is the first slot after slot n belonging to the set $(t_0^{SL}, t_1^{SL}, \dots, t_{T_{max}-1}^{SL})$; otherwise $Q = 1$. T_{scal} is set to selection window size T_2 converted to units of msec.
- 7) If the number of candidate single-slot resources remaining in the set S_A is smaller than $X \cdot M_{total}$, then $Th(p_i, p_j)$ is increased by 3 dB for each priority value $Th(p_i, p_j)$ and the procedure continues with step 4.

TABLE 18

The UE shall report set S_A to higher layers.

If a resource r_i from the set $(r_0, r_1, r_2, \dots)$ is not a member of S_A , then the UE shall report re-evaluation of the resource r_i to higher layers.

If a resource r'_i from the set $(r'_0, r'_1, r'_2, \dots)$ meets the conditions below then the UE shall report pre-emption of the resource r'_i to higher layers

- r'_i is not a member of S_A , and
- r'_i meets the conditions for exclusion in step 6, with $Th(prio_{RX}, prio_{TX})$ set to the final threshold after executing steps 1)-7), i.e. including all necessary increments for reaching $X \cdot M_{total}$ and
- the associated priority $prio_{RX}$, satisfies one of the following conditions:
 - sl-PreemptionEnable is provided and is equal to 'enabled' and $prio_{TX} > prio_{RX}$
 - sl-PreemptionEnable is provided and is not equal to 'enabled', and $prio_{RX} < prio_{pre}$ and $prio_{TX} > prio_{RX}$

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TABLE 19

$T_{proc,0}^{SL}$ depending on sub-carrier spacing	
μ_{SL}	$T_{proc,0}^{SL}$ [slots]
0	1
1	1
2	2
3	4

TABLE 20

$T_{proc,1}^{SL}$ depending on sub-carrier spacing	
μ_{SL}	$T_{proc,1}^{SL}$ [slots]
0	3
1	5
2	9
3	17

Meanwhile, in the existing Uu link-based positioning, a gNB schedules a PRS transmission resource, but in SL positioning, it may be necessary to define an operation for determining an SL PRS transmission resource by a UE based on sensing.

According to an embodiment of the present disclosure, a method for selecting an SL PRS transmission resource based on sensing in order for a UE to determine an SL PRS transmission resource in SL positioning and a device supporting the same are proposed.

For example, for (or, for each of) at least one among elements/parameters of service type (and/or (LCH or service) priority and/or QOS requirements (e.g., latency, reliability, minimum communication range) and/or PQI parameters) (and/or HARQ feedback enabled (and/or disabled) LCH/MAC PDU (transmission) and/or CBR measurement value of a resource pool and/or SL cast type (e.g., unicast, groupcast, broadcast) and/or SL groupcast HARQ feedback option (e.g., NACK only feedback, ACK/NACK feedback, NACK only feedback based on TX-RX distance) and/or SL mode 1 CG type (e.g., SL CG type 1/2) and/or SL mode type (e.g., mode 1/2) and/or resource pool and/or PSFCH resource configured resource pool and/or source (L2) ID (and/or destination (L2) ID) and/or PC5 RRC connection/link and/or SL link and/or (with base station) connection state (e.g., RRC connected state, IDLE state, inactive state) and/or whether an SL HARQ process (ID) and/or (of a transmitting UE or a receiving UE) performs an SL DRX operation and/or whether it is a power saving (transmitting or receiving) UE and/or (from the perspective of a specific UE) case when PSFCH transmission and PSFCH reception (and/or a plurality of PSFCH transmissions (exceeding UE capability)) overlap (and/or a case where PSFCH transmission (and/or PSFCH reception) is omitted) and/or a case where a receiving UE actually (successfully) receives a PSCCH (and/or PSSCH) (re)transmission from a transmitting UE, etc.), whether the rule is applied (and/or the proposed method/rule-related parameter value of the present disclosure) may be specifically (or differently or independently) configured/allowed. In addition, in the present disclosure, “configuration” (or “designation”) wording may be extended and interpreted as a form in which a base station informs a UE through a predefined (physical layer or higher layer) channel/signal (e.g., SIB, RRC, MAC CE) (and/or a form provided through pre-configuration and/or a form in which a UE informs other UEs through a predefined (physi-

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cal layer or higher layer) channel/signal (e.g., SL MAC CE, PC5 RRC)), etc. In addition, in this disclosure, the “PSFCH” wording may be extended and interpreted as “(NR or LTE) PSSCH (and/or (NR or LTE) PSCCH) (and/or (NR or LTE) SL SSB (and/or UL channel/signal))”. And, the methods proposed in the present disclosure may be used in combination with each other (in a new type of manner).

For example, the term “specific threshold” below may refer to a threshold value defined in advance or (pre-) configured by a higher layer (including an application layer) of a network, a base station, or a UE. Hereinafter, the term “specific configuration value” may refer to a value defined in advance or (pre-)configured by a higher layer (including an application layer) of a network, a base station, or a UE. Hereinafter, “configured by a network/base station” may mean an operation in which a base station configures (in advance) a UE by higher layer RRC signaling, configures/signals a UE through MAC CE, or signals a UE through DCI.

In the following disclosure, the following terms are used. UE triggered sidelink (SL) positioning—SL positioning in which the procedure is triggered by a UE.

gNB/location server (LS)/LMF-triggered SL positioning—SL positioning in which the procedure is triggered by a gNB/LS/LMF.

UE-control SL positioning—SL positioning in which an SL positioning group is created by a UE.

gNB-control SL positioning—SL positioning in which an SL positioning group is created by a gNB.

UE-based SL positioning—SL positioning in which the position of a UE is calculated by a UE.

UE-assistance SL positioning—SL positioning in which the position of a UE is calculated by a gNB/LS/LMF.

SL positioning group—UEs participating the SL positioning.

target UE (T-UE)—a UE whose position is calculated.

server UE (S-UE)—a UE assisting in positioning of a T-UE.

LS—location server

MG—a measurement gap in which only an SL PRS transmission is allowed.

MW—a measurement window in which both SL data and an SL PRS can be transmitted in a such way that they are multiplexed.

For example, an SL PRS configuration may include the following information.

- Information related to an SL PRS resource set
 - ID, periodicity, slot offset (slot offset relative to reference timing) of an SL PRS resource set.
 - SL PRS resource repetition factor, time gap (a time interval between SL PRS resources), #symbol (the number of symbols that constituting an SL PRS resource)
 - SL PRS muting option
 - SL PRS resource (a resource on which an SL PRS is transmitted)
- SL PRS resource
 - SL PRS resource ID, slot offset (slot offset relative to the starting time point of an SL PRS resource set), symbol offset (symbol offset relative to the starting time point of a slot)
 - SL PRS sequence ID
 - SL PRS comb size and RE offset
 - (information regarding) SL PRS Quasi co-location (QCL)

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According to an embodiment of the present disclosure, a control channel used for SL PRS transmission (SPCCH) may include the following contents.

- MG index
- SL PRS resource set ID
- SL PRS resource ID
- reserved resource indication (e.g., time resource indication value (TRIV)), frequency RIV (FRIV))
- resource reservation interval (for a periodic/semi-static SL PRS transmission)

For example, the reserved resource indication value may indicate information regarding N SL PRS resources, a corresponding field in an SPCCH may indicate a (RE) resource (the fastest resource in the time and/or frequency domain) with the smallest index in the time and/or frequency domain among resources belonging to each SL PRS resource. Based on the indicated RE resource and SL PRS configuration information configured in an SL resource pool or an MG, the location of an SL PRS resource transmitted by a UE transmitting the SPCCH may be estimated. For example, the reserved resource indication field may indicate the location of a representative resource representing an SL PRS resource instead of a (RE) resource with the minimum index.

For example, the reserved resource indication field may indicate only resources within an MG through which a corresponding SPCCH is transmitted. If an SL PRS resource transmitted in an (MG) region other than the MG in which the corresponding SPCCH is transmitted is indicated, a SPCCH may also signal an MG index to which the SL PRS resource belongs. For example, the reserved resource indication field in an SPCCH transmitted in an MG may signal only information regarding the location of an SL PRS resource belonging to the MG. That is, information regarding the location of an SL PRS resource belonging to an area other than the MG cannot be signaled.

For example, as part of the sensing operation, a UE may measure the RSRP of a reference signal (RS) used in an SPCCH including the SL PRS information.

According to an embodiment of the present disclosure, when only an SL PRS is transmitted without transmission of an SPCCH including control information for the SL PRS in a slot in which the SL PRS is transmitted, as a sensing operation for determining an SP PRS transmission resource, a UE may measure received signal strength indication (RSSI) or RSRP or the correlation between a defined SL PRS resource pattern and a received SL PRS resource pattern for each SL PRS resource pattern, based on an SL PRS configuration configured in the corresponding SL resource pool or MG, for the SL PRS resource patterns (e.g., comb size, RE offset, number of symbols) allowed to be transmitted to the resource pool and/or MG. For example, as described above, a UE may measure RSSI, RSRP, or the correlation based on the SL PRS resource pattern it intends to transmit.

In this case, in the above case, a UE may expect/determine that an SPCCH is transmitted through a separate channel or that an SL PRS is transmitted by reusing an SL PRS resource of the detected position within the next MG or SL PRS resource set.

According to an embodiment of the present disclosure, when an SPCCH is transmitted together with an SL PRS in one slot and only one SL PRS configuration is used in the SL resource pool or MG, resources for which the above-described SPCCH RSRP is greater than or equal to a specific threshold and that satisfy the following conditions may be excluded from resources for transmitting SL PRS resources.

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- an SL PRS resource reserved by an SPCCH
- an SL PRS resource of the periodic transmission location indicated by an SPCCH

According to an embodiment of the present disclosure, when only an SL PRS is transmitted within one slot and only one SL PRS configuration is used within an SL resource pool or MG, based on resources corresponding to the following cases, within the MG or within the SL PRS resource set, SL PRS resources of specific threshold value N corresponding to the same location after the corresponding resource may be excluded from resources for transmitting SL PRS resources.

- A case in which the average RSSI or average RSRP value for all (or some) resources belonging to an SL PRS resource pattern is greater than a specific threshold value for an SL PRS resource pattern for which sensing is performed.

- A case in which a correlation peak value between a resource belonging to an SL PRS resource pattern and an SL PRS resource sequence configured in the resource pool or MG is greater than a specific threshold value for an SL PRS resource pattern for which sensing is performed.

According to an embodiment of the present disclosure, when an SPCCH is transmitted together with an SL PRS in one slot and only a plurality of SL PRS configurations are used in the SL resource pool or MG, if the following conditions are satisfied, SL PRS resources reserved by SPCCH or SL PRS resources of periodic transmission positions indicated by an SPCCH may be excluded from resources for SL PRS transmission to be transmitted by a UE.

- A case in which the sensed SPCCH RSRP is greater than or equal to a specific threshold and the number of overlapping resources between an SL PRS resource pattern that a UE intends to transmit and an PRS resource pattern signaled through an SPCCH is greater than equal to a specific threshold, or the ratio of the number of the overlapped resource compared to the total number of resources included in the SL PRS resource pattern that the UE intends to transmit is greater than or equal to a specific threshold.

According to an embodiment of the present disclosure, when an SPCCH is transmitted together with an SL PRS in one slot and only a plurality of SL PRS configurations are used in the SL resource pool or MG, if the following conditions are satisfied, an SL PRS resource reserved by an SPCCH or an SL PRS resource of a periodic transmission location indicated by an SPCCH may be selected as a resource for SL PRS resource transmission.

- when the sensed SPCCH RSRP is less than or equal to a specific threshold.

- when the sensed SPCCH RSRP is greater than or equal to a specific threshold and the number of overlapping resources between an SL PRS resource pattern that a UE intends to transmit and an PRS resource pattern signaled through an SPCCH is less than or equal to a specific threshold, or the ratio of the number of the overlapped resource compared to the total number of resources included in the SL PRS resource pattern that the UE intends to transmit is less than or equal to a specific threshold.

- when the sensed SPCCH RSRP is greater than or equal to a specific threshold and the number of overlapping resources between an SL PRS resource pattern that a UE intends to transmit and an PRS resource pattern signaled through an SPCCH is less than or equal to a specific threshold, or the ratio of the number of the

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overlapped resource compared to the total number of resources included in the SL PRS resource pattern that the UE intends to transmit is less than or equal to a specific threshold, the SL PRS that the UE intends to transmit may be transmitted using the remaining resources from which only the overlapping resources are excluded among SL PRS resources reserved by the SPCCH or SL PRS resources of the periodic transmission position indicated by the SPCCH.

FIG. 13 shows a procedure for a first UE to transmit an SL PRS based on at least one SL PRS resource according to an embodiment of the present disclosure. The embodiment of FIG. 13 may be combined with various embodiments of the present disclosure.

Referring to FIG. 13, in step S1310, a first UE may determine at least one SL PRS candidate resource for transmitting an SL PRS. In step S1320, the first UE may receive a PRS-related control channel transmitted by a second UE from the second UE. For example, the control channel may include information related to at least one resource that the second UE will use to transmit an SL PRS.

In step S1330, the first UE may update the at least one SL PRS candidate resource, based on information related to at least one resource included in the control channel and to be used by the second UE. For example, the updating may include an operation of excluding at least one resource to be used by the second UE to transmit an SL PRS from the at least one SL PRS candidate resource. For example, the excluded resource may be a resource overlapping with the at least one SL PRS candidate resource.

In step S1340, the first UE may select at least one SL PRS resource to be used for transmitting its own SL PRS among the updated at least one SL PRS candidate resource. And, in step S1350, the first UE may transmit its own SL PRS to a third UE based on the at least one SL PRS resource to be used for transmitting its own SL PRS. For example, positioning for the first UE may be performed based on the SL PRS.

FIG. 14 shows an embodiment in which SL PRS candidate resources are updated according to an embodiment of the present disclosure. The embodiment of FIG. 14 may be combined with various embodiments of the present disclosure.

Referring to FIG. 14, 1410 may represent SL PRS candidate resources selected by a first UE (a UE intending to transmit an SL PRS). Here, the first UE may receive information 1420 related to SL PRS resources related to an SL PRS to be transmitted by a second UE from the second UE.

For example, 1430 represents SL PRS candidate resources in which overlapping resources of 1410 and 1420 are marked. For example, 1440 indicates SL PRS candidate resources from which overlapping resources indicated in 1430 above are excluded.

For example, referring to FIG. 14, a first UE may select SL PRS candidate resources of 1410 first, and may perform an update operation of excluding the overlapping resources marked in 1430 from the SL PRS candidate resources, by receiving SL PRS resource information of a second UE indicated in 1420. For example, finally, the first UE may select an SL PRS resource based on the SL PRS candidate resources of 1440 and transmit its own PRS.

According to various embodiments of the present disclosure, an operation for determining an SL PRS transmission resource based on sensing is proposed according to whether an SL PRS control channel is transmitted together with an SL PRS within one slot in an SL positioning operation.

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Meanwhile, constraints in the time domain to be applied in selecting transmission resources for SL PRS transmission need to be defined.

According to an embodiment of the present disclosure, a method for minimum and maximum time conditions between SL PRS transmission resources to be applied in selecting transmission resources for SL PRS transmission and a device supporting the same are proposed.

SL PRS transmission resources may be composed of a set of SL PRS resources consisting of the following information.

SL PRS resource set ID

SL PRS resource ID list—SL PRS resource ID list in an SL PRS resource set

SL PRS resource type—it may be configured as periodic or aperiodic or semi-persistent or on-demand.

alpha for SL PRS power control

p0 for SL PRS power control

pathloss reference for SL PRS power control—it may be configured as an SL SSB or a DL PRS or a UL SRS or a UL SRS for positioning or a PSCCH DMRS or a PSSCH DMRS or a PSFCH or an SL CSI RS.

The SL PRS resource set may be composed of SL PRS resources composed of the following information.

SL PRS resource ID

SL PRS comb size—an interval between REs in which SL PRSs are transmitted in a symbol

SL PRS comb offset—the RE index in which an SL PRS is first transmitted in the first SL PRS symbol

SL PRS comb cyclic shift—cyclic shift used to generate a sequence that make up an SL PRS

SL PRS start position—the index of the first symbol transmitting an SL PRS in one slot

#symbol of an SL PRS—the number of symbols constituting an SL PRS in one slot

frequency domain shift—the lowest frequency position (index) at which an SL PRS is transmitted in the frequency domain

SL PRS BW—the frequency bandwidth used for an SL PRS transmission

SL PRS resource type—it may be configured as periodic or aperiodic or semi-persistent or on-demand.

SL PRS period—the period in the time domain between SL PRS resources, unit of physical slot or resource pool logical slot in which an SL PRS are transmitted

SL PRS offset—the reference timing reference offset in the time domain up to the start of the first SL PRS resource, unit of physical slot or resource pool logical slot in which an SL PRS are transmitted. The reference timing may be SFN=0 or DFN=0 or a successful reception or decoding time of RRC/MAC-CE/DCI/SCI related to the SL PRS resource.

SL PRS sequence ID

SL PRS spatial relation—it may be configured as an SL SSB or a DL PRS or a UL SRS or a UL SRS for positioning or a PSCCH DMRS or a PSSCH DMRS or a PSFCH or an SL CSI RS.

SL PRS CCH—SL PRS control channel. It may signal SL PRS resource configuration information and the location of a resource, etc.

For example, when performing SL positioning, if UE-B receives an SL PRS transmitted by UE-A and reports a measurement result based on the SL PRS to UE-A,

1. A transmission resource for reporting a measurement result may be located after a time interval greater than or equal to a minimum time gap (min. time gap) (pre-)configured in the resource pool from the time

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point of the SL PRS transmission resource, to ensure that the UE-B secures time necessary for calculating the measurement result based on the SL PRS received from the UE-A.

2. A transmission resource for reporting the measurement result may be located within a maximum time duration (max. time duration) (pre-)configured in a resource pool from the SL PRS transmission resource. For example, when UE-A does not receive the measurement result even after the maximum time interval after the time point at which the SL PRS is transmitted, UE-B may determine that UE-B has not properly received the SL PRS. For example, in the above case, UE-A may perform additional retransmission for the SL PRS.
3. For example, in the above case, when an SL PRS ACK/NACK transmission indicating whether or not an SL PRS is received is configured in the resource pool, if UE-A does not receive an SL PRS ACK/NACK for the SL PRS even after the maximum time interval has elapsed since the SL PRS was transmitted, it may be determined that UE-B did not properly receive the SL PRS. For example, in the above case, UE-A may perform additional retransmission for the SL PRS.
4. For example, the maximum time interval may be indicated to UE-B through an SL PRS CCH related to the SL PRS transmission. For example, the maximum time interval may be determined based on the following items.
 - Target positioning accuracy or target positioning error
 - Latency requirements related to SL positioning related to the SL PRS
 - The priority related to SL positioning related to the SL PRS
 - BW related to a resource pool in which the SL PRS is to be transmitted, the number of logical slots and/or symbols
 - The speed or associated Doppler spread of a UE to select the SL PRS transmission resource
 - The channel delay spread related to the channel/resource pool in which the SL PRS is to be transmitted
 - The congestion level (CBR) related to the channel/resource pool in which the SL PRS is to be transmitted
 - The number of UEs performing SL positioning within the channel/resource pool in which the SL PRS is to be transmitted
 - The distance between UEs to perform positioning by transmitting the SL PRS (e.g., based on zone ID)

According to an embodiment of the present disclosure, when the UE-B is allocated transmission resources from a base station or an LMF, when the UE-B receives the SL PRS CCH from the UE-A, the UE-B may report the maximum time interval information indicated by the SL PRS CCH to a base station.

According to an embodiment of the present disclosure, in the case of UE-assisted SL positioning operation, when UE-A is allocated transmission resources from a base station or LMF, when the base station or the LMF allocates the SL PRS transmission resource to UE-A, for example, the maximum time interval #2 may be configured for UE-A through the DCI allocating the transmission resource. For example, the maximum time interval #2 may be configured to be greater than or equal to the maximum time interval. For example, by selecting the maximum time interval smaller than or equal to the maximum time interval #2 configured by

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the base station or LMF, UE-A may instruct UE-B the selected maximum time interval through the SL PRS CCH transmitted to UE-B.

According to an embodiment of the present disclosure, when UE-A transmits an SL PRS-A to UE-B to perform SL RTT-based positioning, and UE-B transmits an SL PRS-B to UE-A after receiving the SL PRS-A,

1. To secure UE processing time for the UE-B to receive the SL PRS-A received from the UE-A and prepare for a transmission of the SL PRS-B, the SL PRS-A transmission resource and the SL PRS-B transmission resource may be separated by a time interval greater than or equal to a minimum time gap (pre-)configured in a resource pool.
2. The SL PRS-B transmission resource may be located within a maximum time interval (pre-)configured in a resource pool from the SL PRS-A transmission resource. For example, if the SL PRS-B is not received even after the maximum time interval after the SL PRS-A is transmitted, UE-A may determine that UE-B has not properly received the SL PRS-A. For example, in the above case, UE-A may perform additional retransmission for the SL PRS-A.

In the above case, UE-A may release transmission resources for the SL PRS-B, previously reserved and instructed to UE-B. For example, the operation of releasing the SL PRS-B transmission resource may be performed when the time interval between the transmission resource selected for the additional SL PRS-A transmission of UE-A and the transmission resource reserved for the SL PRS-B transmission is smaller than the minimum time gap. For example, in the operation of releasing the SL PRS-B transmission resource, when the time interval between the transmission resources selected for transmission of the additional SL PRS-A of UE-A and the transmission resources reserved for transmission of the SL PRS-B is greater than or equal to the minimum time gap, the previously reserved transmission resource for PRS-B may not be released, and transmission resource information previously reserved for the PRS-B may be included in SL PRS CCH information related to an additional transmission of the SL PRS-A.

In the above case, if the number of (continuous) SL PRS-B reception failures is equal to or greater than a specific threshold, the UE-A may determine that the UE-B is no longer participating in the SL RTT positioning and restart the SL RTT positioning process. That is, the UE-A may transmit an SL RTT positioning request to neighboring UEs and perform the SL RTT positioning process again with UEs that have accepted the request. For example, in the above case, UE-A may release transmission resources previously reserved for the purpose of transmitting the SL PRS-A to UE-B.

3. For example, in the above case, when an SL PRS ACK/NACK transmission indicating whether or not SL PRS has been received is configured in the resource pool, if an SL PRS ACK/NACK for the SL PRS-A is not received even after the maximum time period after the point at which the SL PRS-A is transmitted, UE-A may determine that UE-B did not properly receive the SL PRS-A. For example, in the above case, UE-A may perform additional retransmission for the SL PRS-A.
4. For example, the maximum time interval may be indicated to UE-B through an SL PRS CCH related to

the SL PRS-A transmission. For example, the maximum time interval may be determined based on the following items.

Target positioning accuracy or target positioning error
The latency requirements related to SL positioning related to the SL PRS

The priority related to SL positioning related to the SL PRS

BW related to a resource pool in which the SL PRS is to be transmitted, or BW, the number of logical slots and/or symbols

The speed of a UE to select the SL PRS transmission resource or the related Doppler spread

The channel delay spread related to a channel/resource pool in which the SL PRS is to be transmitted

The congestion level (CBR) related to a channel/resource pool in which the SL PRS is to be transmitted

The number of UEs performing SL positioning within a channel/resource pool in which the SL PRS is to be transmitted.

The distance between UEs to perform positioning by transmitting the SL PRS (e.g., based on zone ID)

According to an embodiment of the present disclosure, when the UE-B is allocated transmission resources from the base station or the LMF, when the UE-B receives the SL PRS CCH from the UE-A, the UE-B may report the maximum time interval information related to the SL PRS-B transmission indicated by the SL PRS CCH to a base station.

According to an embodiment of the present disclosure, in the case of a UE-assisted SL positioning operation, when UE-A is allocated transmission resources from a base station or LMF, when a base station or LMF allocates the SL PRS transmission resource to UE-A, for example, through DCI allocating the transmission resource, the maximum time interval #2 related to the SL PRS-B transmission may be configured for UE-A. For example, the maximum time interval #2 related to the SL PRS-B transmission may be configured to be greater than or equal to the maximum time interval related to the SL PRS-B transmission. For example, by selecting the maximum time interval related to the SL PRS-B transmission that is less than or equal to the maximum time interval #2 related to the SL PRS-B transmission configured by the base station or the LMF, UE-A may indicate the selected maximum time interval to UE-B through the SL PRS CCH transmitted to UE-B.

According to an embodiment of the present disclosure, after receiving an SL PRS CCH related to the SL PRS-A from the UE-A, if the SL PRS-A is not received within a specific threshold time (pre-)configured in the resource pool, the UE-B may transmit an SL PRS NACK for the SL PRS-A to UE-A.

According to various embodiments of the present disclosure, by setting minimum and maximum time conditions between SL PRS transmission resources to be applied to select a transmission resource for SL PRS transmission, an efficient resource management method was proposed by securing UE processing time for SL PRS processing and checking whether SL PRS was received.

According to the existing technology, positioning between UEs performing SL communication could not be performed. According to an embodiment of the present disclosure, based on an SL PRS, positioning between UEs performing SL communication may be performed, and when selecting an SL PRS resource related to transmission of the

SL PRS, positioning can be performed more efficiently by excluding resources used by other UEs to transmit an SL PRS.

FIG. 15 shows a procedure for performing wireless communication by a first device according to an embodiment of the present disclosure. The embodiment of FIG. 15 may be combined with various embodiments of the present disclosure.

Referring to FIG. 15, in step S1510, a first device may determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS. In step S1520, the first device may receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS. In step S1530, the first device may obtain a reference signal received power (RSRP) value of the first control channel. In step S1540, the first device may obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information. In step S1550, the first device may exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold. In step S1560, the first device may transmit the first SL PRS based on the at least one first SL PRS candidate resource.

For example, the first SL PRS may be transmitted based on the at least one first SL PRS candidate resource, in which the at least one second SL PRS resource is excluded.

For example, additionally, the first device may obtain information related to a measurement gap period in which a plurality of sidelink (SL) positioning reference signal (PRS) configurations are configured. For example, a transmission of SL data in the measurement gap period may be not allowed.

For example, additionally, the first device may perform positioning for the first device based on the first SL PRS.

For example, the at least one second SL PRS resource may be not excluded from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to the first threshold and the number of the at least one second SL PRS resource being less than or equal to the second threshold.

For example, the at least one first SL PRS candidate resource and the at least one second SL PRS resource may be pattern forms.

For example, the at least one first SL PRS candidate resource may be determined based on a first SL PRS configuration, and the at least one second SL PRS resource may be determined based on a second SL PRS configuration.

For example, the first control information may include information related to a measurement gap index.

For example, the first control information may include information related to the at least one first SL PRS resource.

For example, the first control information may include information related to a resource with a lowest index among the at least one first SL PRS resource.

For example, the first control information may include information regarding a location of a representative resource of the at least one first SL PRS resource.

For example, the first control information may include an SL PRS resource ID related to the at least one first SL PRS resource.

For example, the first control information may include an SL PRS resource set ID related to the at least one first SL PRS resource.

The above-described embodiment may be applied to various devices described below. First, a processor 102 of a first device 100 may determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS. And, the processor 102 of a first device 100 may control a transceiver 106 to receive, from a second device 200, first control information related to SL PRS through a first control channel related to SL PRS. And, the processor 102 of a first device 100 may obtain a reference signal received power (RSRP) value of the first control channel. And, the processor 102 of a first device 100 may obtain information related to at least one first SL PRS resource to be used by the second device 200, based on the first control information. And, the processor 102 of a first device 100 may exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold. And, the processor 102 of a first device 100 may control the transceiver 106 to transmit the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a first device for performing wireless communication may be proposed. For example, the first device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers. For example, the one or more processors may execute the instructions to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

For example, the first SL PRS may be transmitted based on the at least one first SL PRS candidate resource, in which the at least one second SL PRS resource is excluded.

For example, additionally, the first device may obtain information related to a measurement gap period in which a plurality of sidelink (SL) positioning reference signal (PRS) configurations are configured. For example, a transmission of SL data in the measurement gap period may be not allowed.

For example, additionally, the first device may perform positioning for the first device based on the first SL PRS.

For example, the at least one second SL PRS resource may be not excluded from the at least one first SL PRS candidate resource, based on the RSRP value being greater

than or equal to the first threshold and the number of the at least one second SL PRS resource being less than or equal to the second threshold.

For example, the at least one first SL PRS candidate resource and the at least one second SL PRS resource may be pattern forms.

For example, the at least one first SL PRS candidate resource may be determined based on a first SL PRS configuration, and the at least one second SL PRS resource may be determined based on a second SL PRS configuration.

For example, the first control information may include information related to a measurement gap index.

For example, the first control information may include information related to the at least one first SL PRS resource.

For example, the first control information may include information related to a resource with a lowest index among the at least one first SL PRS resource.

For example, the first control information may include information regarding a location of a representative resource of the at least one first SL PRS resource.

For example, the first control information may include an SL PRS resource ID related to the at least one first SL PRS resource.

For example, the first control information may include an SL PRS resource set ID related to the at least one first SL PRS resource.

According to an embodiment of the present disclosure, a device adapted to control a first user equipment (UE), the device comprising: one or more processors; and one or more memories operably connectable to the one or more processors and storing instructions may be proposed. For example, the one or more processors may execute the instructions to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second UE, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second UE, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

According to an embodiment of the present disclosure, a non-transitory computer-readable storage medium storing instructions may be proposed. For example, the instructions, when executed, may cause a first device to: determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS; receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS; obtain a reference signal received power (RSRP) value of the first control channel; obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information; exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second

threshold; and transmit the first SL PRS based on the at least one first SL PRS candidate resource.

FIG. 16 shows a procedure for performing wireless communication by a second device according to an embodiment of the present disclosure. The embodiment of FIG. 16 may be combined with various embodiments of the present disclosure.

Referring to FIG. 16, in step S1610, a second device may receive, from a first device, a first positioning reference signal (PRS) based on at least one first sidelink (SL) PRS resource. In step S1620, the second device may transmit, to the first device, a second PRS based on the reception of the first PRS. For example, first control information related to PRS may be received to the first device through a first control channel related to PRS, a third SL PRS resource may be excluded from at least one first SL PRS candidate resource, based on an RSRP value of the first control channel being greater than or equal to a first threshold and a number of the at least one third SL PRS resource, in which the at least one first SL PRS candidate resource and at least one second SL PRS resource overlap, being greater than or equal to a second threshold, information related to the at least one second SL PRS resource may be obtained by the first device based on the first control information, the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, and positioning for the first device may be performed based on the first PRS and the second PRS.

For example, the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, in which the at least one third PRS resource is excluded.

The above-described embodiment may be applied to various devices described below. First, a processor 202 of a second device 200 may control a transceiver 206 to receive, from a first device 100, a first positioning reference signal (PRS) based on at least one first sidelink (SL) PRS resource. And, the processor 202 of the second device 200 may control the transceiver 206 transmit, to the first device 100, a second PRS based on the reception of the first PRS. For example, first control information related to PRS may be received to the first device 100 through a first control channel related to PRS, a third SL PRS resource may be excluded from at least one first SL PRS candidate resource, based on an RSRP value of the first control channel being greater than or equal to a first threshold and a number of the at least one third SL PRS resource, in which the at least one first SL PRS candidate resource and at least one second SL PRS resource overlap, being greater than or equal to a second threshold, information related to the at least one second SL PRS resource may be obtained by the first device 100 based on the first control information, the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, and positioning for the first device 100 may be performed based on the first PRS and the second PRS.

According to an embodiment of the present disclosure, a second device for performing wireless communication may be proposed. For example, the second device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers, wherein the one or more processors execute the instructions to: receive, from a first device, a first positioning reference signal (PRS) based on at least one first sidelink (SL) PRS resource; and transmit, to the first device, a second PRS based on the reception of the first PRS, wherein first control

information related to PRS may be received to the first device through a first control channel related to PRS, wherein a third SL PRS resource may be excluded from at least one first SL PRS candidate resource, based on an RSRP value of the first control channel being greater than or equal to a first threshold and a number of the at least one third SL PRS resource, in which the at least one first SL PRS candidate resource and at least one second SL PRS resource overlap, being greater than or equal to a second threshold, wherein information related to the at least one second SL PRS resource may be obtained by the first device based on the first control information, wherein the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, and wherein positioning for the first device may be performed based on the first PRS and the second PRS.

For example, the at least one first SL PRS resource may be selected among the at least one first SL PRS candidate resource, in which the at least one third PRS resource is excluded.

Various embodiments of the present disclosure may be combined with each other.

Hereinafter, device(s) to which various embodiments of the present disclosure can be applied will be described.

The various descriptions, functions, procedures, proposals, methods, and/or operational flowcharts of the present disclosure described in this document may be applied to, without being limited to, a variety of fields requiring wireless communication/connection (e.g., 5G) between devices.

Hereinafter, a description will be given in more detail with reference to the drawings. In the following drawings/description, the same reference symbols may denote the same or corresponding hardware blocks, software blocks, or functional blocks unless described otherwise.

FIG. 17 shows a communication system 1, based on an embodiment of the present disclosure. The embodiment of FIG. 17 may be combined with various embodiments of the present disclosure.

Referring to FIG. 17, a communication system 1 to which various embodiments of the present disclosure are applied includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an eXtended Reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of Things (IoT) device 100f, and an Artificial Intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smart-pad, a wearable device (e.g., a smartwatch or a smart-glasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be

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implemented as wireless devices and a specific wireless device **200a** may operate as a BS/network node with respect to other wireless devices.

Here, wireless communication technology implemented in wireless devices **100a** to **100f** of the present disclosure may include Narrowband Internet of Things for low-power communication in addition to LTE, NR, and 6G. In this case, for example, NB-IoT technology may be an example of Low Power Wide Area Network (LPWAN) technology and may be implemented as standards such as LTE Cat NB1, and/or LTE Cat NB2, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100a** to **100f** of the present disclosure may perform communication based on LTE-M technology. In this case, as an example, the LTE-M technology may be an example of the LPWAN and may be called by various names including enhanced Machine Type Communication (eMTC), and the like. For example, the LTE-M technology may be implemented as at least any one of various standards such as 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-Bandwidth Limited (non-BL), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100a** to **100f** of the present disclosure may include at least one of Bluetooth, Low Power Wide Area Network (LPWAN), and ZigBee considering the low-power communication, and is not limited to the name described above. As an example, the ZigBee technology may generate personal area networks (PAN) related to small/low-power digital communication based on various standards including IEEE 802.15.4, and the like, and may be called by various names.

The wireless devices **100a** to **100f** may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices **100a** to **100f** may communicate with each other through the BSs **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

Wireless communication/connections **150a**, **150b**, or **150c** may be established between the wireless devices **100a** to **100f**/BS **200**, or BS **200**/BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150a**, sidelink communication **150b** (or, D2D communication), or inter BS communication (e.g. relay, Integrated Access Backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150a** and **150b**. For example, the wireless communication/connections **150a** and **150b** may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodula-

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tion, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

FIG. **18** shows wireless devices, based on an embodiment of the present disclosure. The embodiment of FIG. **18** may be combined with various embodiments of the present disclosure.

Referring to FIG. **18**, a first wireless device **100** and a second wireless device **200** may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device **100** and the second wireless device **200**} may correspond to {the wireless device **100x** and the BS **200**} and/or {the wireless device **100x** and the wireless device **100x**} of FIG. **17**.

The first wireless device **100** may include one or more processors **102** and one or more memories **104** and additionally further include one or more transceivers **106** and/or one or more antennas **108**. The processor(s) **102** may control the memory(s) **104** and/or the transceiver(s) **106** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **102** may process information within the memory(s) **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) **106**. The processor(s) **102** may receive radio signals including second information/signals through the transceiver **106** and then store information obtained by processing the second information/signals in the memory(s) **104**. The memory(s) **104** may be connected to the processor(s) **102** and may store a variety of information related to operations of the processor(s) **102**. For example, the memory(s) **104** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **102** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **102** and the memory(s) **104** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **106** may be connected to the processor(s) **102** and transmit and/or receive radio signals through one or more antennas **108**. Each of the transceiver(s) **106** may include a transmitter and/or a receiver. The transceiver(s) **106** may be interchangeably used with Radio Frequency (RF) unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

The second wireless device **200** may include one or more processors **202** and one or more memories **204** and additionally further include one or more transceivers **206** and/or one or more antennas **208**. The processor(s) **202** may control the memory(s) **204** and/or the transceiver(s) **206** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **202** may process information within the memory(s) **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) **206**. The processor(s) **202** may receive radio signals including fourth information/signals through the transceiver(s) **106** and then store information obtained by processing the fourth information/signals in the memory(s) **204**. The memory(s) **204** may be connected to the processor(s) **202** and may store a variety of information related to operations of the processor(s) **202**. For example, the memory(s) **204** may store software code including

commands for performing a part or the entirety of processes controlled by the processor(s) **202** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **202** and the memory(s) **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **206** may be connected to the processor(s) **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver(s) **206** may include a transmitter and/or a receiver. The transceiver(s) **206** may be interchangeably used with RF unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors **102** and **202** may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this

document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

FIG. **19** shows a signal process circuit for a transmission signal, based on an embodiment of the present disclosure. The embodiment of FIG. **19** may be combined with various embodiments of the present disclosure.

Referring to FIG. **19**, a signal processing circuit **1000** may include scramblers **1010**, modulators **1020**, a layer mapper **1030**, a precoder **1040**, resource mappers **1050**, and signal generators **1060**. An operation/function of FIG. **19** may be performed, without being limited to, the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. **18**. Hardware elements of FIG. **19** may be implemented by the

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processors **102** and **202** and/or the transceivers **106** and **206** of FIG. **18**. For example, blocks **1010** to **1060** may be implemented by the processors **102** and **202** of FIG. **18**. Alternatively, the blocks **1010** to **1050** may be implemented by the processors **102** and **202** of FIG. **18** and the block **1060** may be implemented by the transceivers **106** and **206** of FIG. **18**.

Codewords may be converted into radio signals via the signal processing circuit **1000** of FIG. **19**. Herein, the codewords are encoded bit sequences of information blocks. The information blocks may include transport blocks (e.g., a UL-SCH transport block, a DL-SCH transport block). The radio signals may be transmitted through various physical channels (e.g., a PUSCH and a PDSCH).

Specifically, the codewords may be converted into scrambled bit sequences by the scramblers **1010**. Scramble sequences used for scrambling may be generated based on an initialization value, and the initialization value may include ID information of a wireless device. The scrambled bit sequences may be modulated to modulation symbol sequences by the modulators **1020**. A modulation scheme may include pi/2-Binary Phase Shift Keying (pi/2-BPSK), m-Phase Shift Keying (m-PSK), and m-Quadrature Amplitude Modulation (m-QAM). Complex modulation symbol sequences may be mapped to one or more transport layers by the layer mapper **1030**. Modulation symbols of each transport layer may be mapped (precoded) to corresponding antenna port(s) by the precoder **1040**. Outputs z of the precoder **1040** may be obtained by multiplying outputs y of the layer mapper **1030** by an $N \times M$ precoding matrix W . Herein, N is the number of antenna ports and M is the number of transport layers. The precoder **1040** may perform precoding after performing transform precoding (e.g., DFT) for complex modulation symbols. Alternatively, the precoder **1040** may perform precoding without performing transform precoding.

The resource mappers **1050** may map modulation symbols of each antenna port to time-frequency resources. The time-frequency resources may include a plurality of symbols (e.g., a CP-OFDM symbols and DFT-s-OFDM symbols) in the time domain and a plurality of subcarriers in the frequency domain. The signal generators **1060** may generate radio signals from the mapped modulation symbols and the generated radio signals may be transmitted to other devices through each antenna. For this purpose, the signal generators **1060** may include Inverse Fast Fourier Transform (IFFT) modules, Cyclic Prefix (CP) inserters, Digital-to-Analog Converters (DACs), and frequency up-converters.

Signal processing procedures for a signal received in the wireless device may be configured in a reverse manner of the signal processing procedures **1010** to **1060** of FIG. **19**. For example, the wireless devices (e.g., **100** and **200** of FIG. **18**) may receive radio signals from the exterior through the antenna ports/transceivers. The received radio signals may be converted into baseband signals through signal restorers. To this end, the signal restorers may include frequency downlink converters, Analog-to-Digital Converters (ADCs), CP remover, and Fast Fourier Transform (FFT) modules. Next, the baseband signals may be restored to codewords through a resource demapping procedure, a postcoding procedure, a demodulation processor, and a descrambling procedure. The codewords may be restored to original information blocks through decoding. Therefore, a signal processing circuit (not illustrated) for a reception signal may include signal restorers, resource demappers, a postcoder, demodulators, descramblers, and decoders.

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FIG. **20** shows another example of a wireless device, based on an embodiment of the present disclosure. The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. **17**). The embodiment of FIG. **20** may be combined with various embodiments of the present disclosure.

Referring to FIG. **20**, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. **18** and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. **18**. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. **18**. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. **17**), the vehicles (**100b-1** and **100b-2** of FIG. **17**), the XR device (**100c** of FIG. **17**), the hand-held device (**100d** of FIG. **17**), the home appliance (**100e** of FIG. **17**), the IoT device (**100f** of FIG. **17**), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. **17**), the BSs (**200** of FIG. **17**), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. **20**, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a Random

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Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM)), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

Hereinafter, an example of implementing FIG. 20 will be described in detail with reference to the drawings.

FIG. 21 shows a hand-held device, based on an embodiment of the present disclosure. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT). The embodiment of FIG. 21 may be combined with various embodiments of the present disclosure.

Referring to FIG. 21, a hand-held device 100 may include an antenna unit 108, a communication unit 110, a control unit 120, a memory unit 130, a power supply unit 140a, an interface unit 140b, and an I/O unit 140c. The antenna unit 108 may be configured as a part of the communication unit 110. Blocks 110 to 130/140a to 140c correspond to the blocks 110 to 130/140 of FIG. 20, respectively.

The communication unit 110 may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit 120 may perform various operations by controlling constituent elements of the hand-held device 100. The control unit 120 may include an Application Processor (AP). The memory unit 130 may store data/parameters/programs/code/commands needed to drive the hand-held device 100. The memory unit 130 may store input/output data/information. The power supply unit 140a may supply power to the hand-held device 100 and include a wired/wireless charging circuit, a battery, etc. The interface unit 140b may support connection of the hand-held device 100 to other external devices. The interface unit 140b may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit 140c may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit 140c may include a camera, a microphone, a user input unit, a display unit 140d, a speaker, and/or a haptic module.

As an example, in the case of data communication, the I/O unit 140c may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit 130. The communication unit 110 may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit 110 may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit 130 and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit 140c.

FIG. 22 shows a vehicle or an autonomous vehicle, based on an embodiment of the present disclosure. The vehicle or autonomous vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned Aerial Vehicle (AV), a ship, etc. The embodiment of FIG. 22 may be combined with various embodiments of the present disclosure.

Referring to FIG. 22, a vehicle or autonomous vehicle 100 may include an antenna unit 108, a communication unit 110, a control unit 120, a driving unit 140a, a power supply unit 140b, a sensor unit 140c, and an autonomous driving unit 140d. The antenna unit 108 may be configured as a part of

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the communication unit 110. The blocks 110/130/140a to 140d correspond to the blocks 110/130/140 of FIG. 20, respectively.

The communication unit 110 may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit 120 may perform various operations by controlling elements of the vehicle or the autonomous vehicle 100. The control unit 120 may include an Electronic Control Unit (ECU). The driving unit 140a may cause the vehicle or the autonomous vehicle 100 to drive on a road. The driving unit 140a may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit 140b may supply power to the vehicle or the autonomous vehicle 100 and include a wired/wireless charging circuit, a battery, etc. The sensor unit 140c may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit 140c may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit 140d may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

For example, the communication unit 110 may receive map data, traffic information data, etc. from an external server. The autonomous driving unit 140d may generate an autonomous driving path and a driving plan from the obtained data. The control unit 120 may control the driving unit 140a such that the vehicle or the autonomous vehicle 100 may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit 110 may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit 140c may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit 140d may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit 110 may transfer information regarding a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous vehicles and provide the predicted traffic information data to the vehicles or the autonomous vehicles.

Claims in the present description can be combined in a various way. For instance, technical features in method claims of the present description can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims can be combined to be implemented or performed in a method. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in an apparatus. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in a method.

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What is claimed is:

1. A method for performing, by a first device, wireless communication, the method comprising:
 - determining at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS;
 - receiving, from a second device, first control information related to SL PRS through a first control channel related to SL PRS;
 - obtaining a reference signal received power (RSRP) value of the first control channel;
 - obtaining information related to at least one first SL PRS resource to be used by the second device, based on the first control information;
 - excluding at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and
 - transmitting the first SL PRS based on the at least one first SL PRS candidate resource.
2. The method of claim 1, wherein the first SL PRS is transmitted based on the at least one first SL PRS candidate resource, in which the at least one second SL PRS resource is excluded.
3. The method of claim 1, further comprising:
 - obtaining information related to a measurement gap period in which a plurality of sidelink (SL) positioning reference signal (PRS) configurations are configured, wherein a transmission of SL data in the measurement gap period is not allowed.
4. The method of claim 1, further comprising:
 - performing positioning for the first device based on the first SL PRS.
5. The method of claim 1, wherein the at least one second SL PRS resource is not excluded from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to the first threshold and the number of the at least one second SL PRS resource being less than or equal to the second threshold.
6. The method of claim 1, wherein the at least one first SL PRS candidate resource and the at least one second SL PRS resource are pattern forms.
7. The method of claim 1, wherein the at least one first SL PRS candidate resource is determined based on a first SL PRS configuration, and
 - wherein the at least one second SL PRS resource is determined based on a second SL PRS configuration.
8. The method of claim 1, wherein the first control information includes information related to a measurement gap index.
9. The method of claim 1, wherein the first control information includes information related to the at least one first SL PRS resource.
10. The method of claim 9, wherein the first control information includes information related to a resource with a lowest index among the at least one first SL PRS resource.
11. The method of claim 9, wherein the first control information includes information regarding a location of a representative resource of the at least one first SL PRS resource.

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12. The method of claim 9, wherein the first control information includes an SL PRS resource ID related to the at least one first SL PRS resource.
13. The method of claim 9, wherein the first control information includes an SL PRS resource set ID related to the at least one first SL PRS resource.
14. A first device for performing wireless communication, the first device comprising:
 - one or more memories storing instructions;
 - one or more transceivers; and
 - one or more processors connected to the one or more memories and the one or more transceivers, wherein the one or more processors execute the instructions to:
 - determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS;
 - receive, from a second device, first control information related to SL PRS through a first control channel related to SL PRS;
 - obtain a reference signal received power (RSRP) value of the first control channel;
 - obtain information related to at least one first SL PRS resource to be used by the second device, based on the first control information;
 - exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and
 - transmit the first SL PRS based on the at least one first SL PRS candidate resource.
15. A device adapted to control a first user equipment (UE), the device comprising:
 - one or more processors; and
 - one or more memories operably connectable to the one or more processors and storing instructions, wherein the one or more processors execute the instructions to:
 - determine at least one first SL positioning reference signal (PRS) candidate resource for transmitting a first PRS;
 - receive, from a second UE, first control information related to SL PRS through a first control channel related to SL PRS;
 - obtain a reference signal received power (RSRP) value of the first control channel;
 - obtain information related to at least one first SL PRS resource to be used by the second UE, based on the first control information;
 - exclude at least one second SL PRS resource from the at least one first SL PRS candidate resource, based on the RSRP value being greater than or equal to a first threshold and a number of the at least one second SL PRS resource, in which the at least one first SL PRS candidate resource and the at least one first SL PRS resource overlap, being greater than or equal to a second threshold; and
 - transmit the first SL PRS based on the at least one first SL PRS candidate resource.

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